

Avoidance of hydrogen sulfide is modulated by external and internal states in *C. elegans*

Reviewed Preprint

v2 • September 12, 2025

Revised by authors

Reviewed Preprint

v1 • December 27, 2023

Longjun Pu, Lina Zhao, Jing Wang, Clementine Deleuze, Lars Nilsson, Johan Henriksson, Patrick Laurent , Changchun Chen 

Department of Molecular Biology, Umeå University, Umeå, Sweden • Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden • Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden • Laboratory of Neurophysiology, ULB Neuroscience Institute (UNI), Université Libre de Bruxelles (ULB), Brussels Belgium • The Laboratory for Molecular Infection Medicine Sweden (MIMS), Umeå University, Umeå, Sweden • Integrated Science Lab (Icelab), Umeå University, Umeå, Sweden

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This paper provides **valuable** insights into the consequences of hydrogen sulfide (H₂S) exposure on the behavior and physiology of the nematode *C. elegans*. While **solid** evidence supports most of the paper's findings, the evidence that H₂S is detected by the nervous system to mediate behavioral avoidance is **incomplete**. The paper provides a wide range of intriguing observations that could serve as a foundation for future work to synthesize these disparate results or provide insight into the mechanisms of H₂S detection in *C. elegans*.

<https://doi.org/10.7554/eLife.92964.2.sa3>

Abstract

Hydrogen sulfide (H₂S) acts as an energy source, a toxin, and a gasotransmitter across diverse biological contexts. We use the robust locomotory responses of *Caenorhabditis elegans* to high levels of H₂S to elucidate the molecular mechanisms underlying its acute perception and adaptive responses. We find that the H₂S-evoked behavioral response is shaped by multiple environmental factors including oxygen (O₂) levels and nutritional state, and is modulated by various pathways such as insulin, TGF-β, and HIF-1 signaling, as well as by input from O₂-sensing neurons. Prolonged exposure to H₂S activates HIF-1 signaling, leading to the upregulation of stress-responsive genes, including those involved in H₂S detoxification. This promotes an adaptive state in which locomotory speed is reduced in H₂S, while responsiveness to other stimuli is preserved. In mutants deficient in HIF-1 signaling, iron storage, and detoxification mechanisms, animals display a robust initial response but rapidly enter a sleep-like behavior characterized by reduced mobility and diminished responsiveness to subsequent sensory stimuli. Furthermore, while acute production of mitochondria-derived reactive O₂ species (ROS) appears to initiate the avoidance response to H₂S, persistently high ROS promotes an adaptive state, likely by activating various stress-response pathways,

without substantially compromising cellular H₂S detoxification capacity. Taken together, our study provides comprehensive molecular insights into the mechanisms through which *C. elegans* detects, modulates, and adapts its response to H₂S exposure.

Introduction

During the late Proterozoic era, the rise in oxygen (O₂) levels eliminated hydrogen sulfide (H₂S) from most habitats. Yet, low O₂ levels, along with high concentrations of hydrogen sulfide (H₂S) and carbon dioxide (CO₂), can persist in enclosed environments where bacteria actively decompose organic matter (Olson & Straub, 2016 [↗](#)). H₂S can easily permeate biological membranes and interfere with various cellular processes. One of the most detrimental effects of H₂S is the disruption of cellular respiration by remodeling the mitochondrial electron transport chain and inhibiting cytochrome c oxidase (COX) (Cooper & Brown, 2008 [↗](#); Khan et al., 1990 [↗](#); Nicholls & Kim, 1982 [↗](#); Romanelli-Cedrez, Vairoletti, & Salinas, 2024 [↗](#)).

The detection and response to chemical signals are crucial for various organisms to interact effectively with their environment. Animals living in H₂S-rich environments have evolved mechanisms to either avoid or adapt to these conditions, conferring them survival advantages. Species that encounter periodic increases in H₂S often possess sensing abilities and exhibit avoidance behaviors. For example, benthic species undertake daily vertical migrations between sulfidic and non-sulfidic waters (Abel, Koenig, & Davis, 1987 [↗](#); Salvanes, Utne-Palm, Currie, & Braithwaite, 2011 [↗](#)). Certain metazoan species have seized the ecological opportunities presented by H₂S-rich environments through the development of cellular adaptations. Livebearing fish species, for instance, have evolved COX subunits that are more resistant to H₂S and more efficient at H₂S detoxification (Kelley et al., 2016 [↗](#); Pfenninger et al., 2014 [↗](#)).

Free-living nematodes feed on bacteria that thrive in decaying organic matter, such as compost. This complex and dynamic environment, where species like *C. elegans* are commonly found, may exhibit significant fluctuations in H₂S levels over short distances (Adams, Farwell, Pack, & Barnesberger, 1979 [↗](#); Budde & Roth, 2011 [↗](#); Morra & Dick, 1991 [↗](#); Patange, Breen, Arsuffi, & Ruvkun, 2025 [↗](#); Rodriguez-Kabana, Jordan, & Hollis, 1965 [↗](#); Romanelli-Cedrez et al., 2024 [↗](#)). Evidence suggests that *C. elegans* has evolved tolerance to prolonged H₂S exposure by reprogramming gene expression through H₂S-induced stabilization of the hypoxia-inducible factor HIF-1 (Budde & Roth, 2010 [↗](#); Powell-Coffman, 2010 [↗](#)), and by developing an H₂S-resistant electron transport chain (Romanelli-Cedrez et al., 2024 [↗](#)). In particular, low concentrations of H₂S are not only well tolerated but can also be beneficial. For instance, exposure to 50 ppm H₂S has been shown to extend the lifespan and enhance thermotolerance (Budde & Roth, 2010 [↗](#); Fawcett, Hoyt, Johnson, & Miller, 2015 [↗](#); Horsman, Heinis, & Miller, 2019 [↗](#); Miller & Roth, 2007 [↗](#); Qabazard et al., 2014 [↗](#)). Moreover, endogenous H₂S production in *C. elegans* modulates its interactions with actinobacteria (Patange et al., 2025 [↗](#)). However, exposure to concentrations exceeding 150 ppm proved lethal (Budde & Roth, 2010 [↗](#), 2011 [↗](#); Fawcett et al., 2015 [↗](#); Horsman et al., 2019 [↗](#); Miller, Budde, & Roth, 2011a [↗](#); Miller & Roth, 2007 [↗](#)). In addition to long-term adaptive mechanisms, *C. elegans* exhibits an acute behavioral response to H₂S exposure (Budde & Roth, 2011 [↗](#)). In this study, we investigated whether and how the nematode *C. elegans* acutely avoids environments rich in H₂S, how it adapts at the molecular and physiological levels to H₂S exposure, and how these adaptations alter the avoidance response.

Results

C. elegans increases locomotory activity in response to acute H₂S exposure

In response to aversive cues, *C. elegans* initiates a pirouette followed by an increase of its locomotory speed to escape the noxious stimuli. When acutely exposed to 150 ppm H₂S, the laboratory N2 strain exhibited an increased locomotory speed and enhanced turning behavior, while generally decreasing the frequency of reversals, as a strategy to escape the noxious stimuli (**Figure 1A–C**, **Figure 1—Video 1**). The maximum speed was reached after 6–8 minutes in H₂S and returned to the baseline over a period of approximately 1 hour (**Figure 1D**). The reduced locomotory speed after 30 minutes in 150 ppm H₂S was reversible when these animals were immediately challenged with another stimulus, such as near-UV light (**Figure 1E**), suggesting that neuromuscular function remains preserved. The magnitude and dynamics of the H₂S-induced avoidance responses were concentration-dependent. Under our assay conditions, 50 ppm H₂S did not elicit an acute speed increase, and responses to 75 ppm were variable. In contrast, exposure to 150 ppm H₂S consistently triggered a robust and reproducible escape response (**Figure 1F**), consistent with earlier reports that H₂S at 50 ppm exerts beneficial effects, while H₂S at 150 ppm is toxic to *C. elegans* (Budde & Roth, 2010; Fawcett et al., 2015; Horsman et al., 2019; Miller & Roth, 2007; Qabazard et al., 2014). To investigate whether low levels of H₂S might be preferable or even attractive to the animals, we exposed wild type animals to a linear gradient of H₂S (**Figure 1G**; See methods for details) (Bretscher, Busch, & de Bono, 2008; Chang, Chronis, Karow, Marletta, & Bargmann, 2006; Gray et al., 2004). Animals were allowed to move freely on a thin layer of bacteria within a microfluidic chamber containing an H₂S gradient ranging from 150 ppm at one end to 0 ppm at the other. As expected, *C. elegans* robustly avoids 150 ppm H₂S but does not accumulate at the extreme end of 0 ppm (**Figure 1G, H**). The highest proportion of animals was found in the region with approximately 40 ppm H₂S. These data suggest that H₂S acts as a potent repellent for *C. elegans* at high concentrations but as an attractant at low levels.

Candidate gene screen identifies signaling pathways required for H₂S-evoked locomotion

To investigate the molecular mechanisms underlying the H₂S-evoked locomotory response in *C. elegans*, we conducted a candidate gene screen for mutants that failed to increase their locomotory speed upon exposure to an acute rise of H₂S level to 150 ppm. Our analysis prioritized genes that are either known to be, or potentially involved in, sensory responses to gaseous stimuli in both *C. elegans* and mammals (**Supplementary file 1**). We began by examining mutants affecting globins, guanylate cyclases, and cyclic nucleotide-gated (CNG) channels, which are involved in O₂ or CO₂ sensing in *C. elegans*, as well as wild isolates and other previously characterized mutants with defective responses to gas stimuli (**Supplementary file 1**) (Bretscher et al., 2008; Gray et al., 2004; Hallem & Sternberg, 2008; Persson et al., 2009). Given the essential role of K⁺ channels in acute hypoxia sensing in mammals, we also screened all potassium (K⁺) channel mutants (Gao, Gonzalez-Rodriguez, Ortega-Saenz, & Lopez-Barneo, 2017; Weir, Lopez-Barneo, Buckler, & Archer, 2005). To explore how the H₂S sensory signals were transmitted within the neural circuit, we assayed mutants with defects in ciliogenesis or cilia-mediated signaling, as well as animals deficient in neurotransmission involving classical neurotransmitters, neuropeptides, or biogenic amines. Finally, considering that mitochondrial function is central to H₂S detoxification, we included mitochondrial electron transport chain (ETC) mutants and key factors implicated in H₂S and superoxide clearance. Together, these functionally diverse categories allowed us to survey a wide spectrum of potential mechanisms contributing to the behavioral response to H₂S.

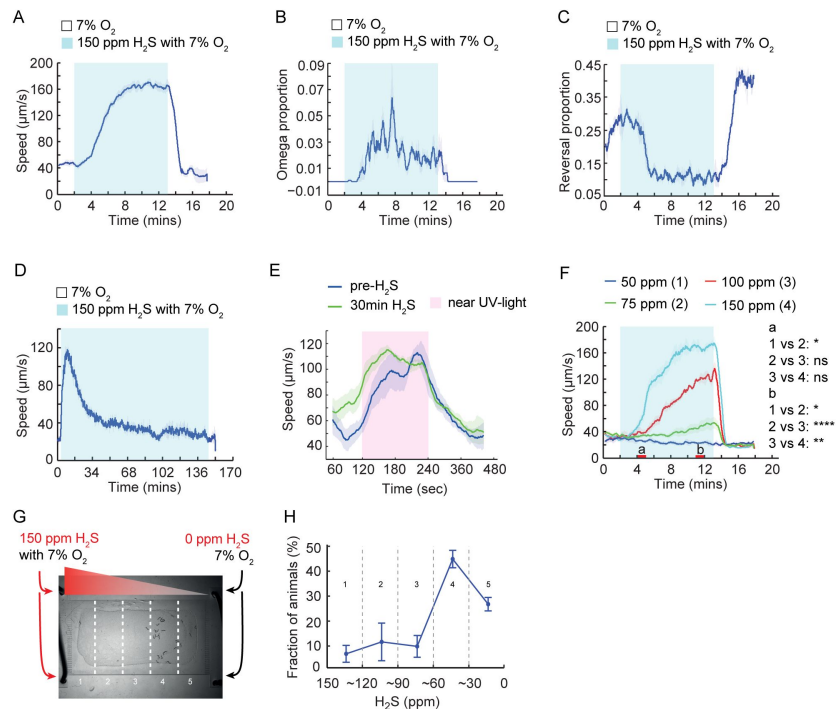


Figure 1.

Acute locomotory responses of *C. elegans* to hydrogen sulfide

(A) Changes in the locomotory speed of WT animals (N2 laboratory strain) evoked by a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂, followed by a return to 7% O₂. (B) Proportion of WT animals undergoing reorientation movements (omega turns) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂, followed by a return to 7% O₂. (C) Fraction of WT animals doing backward locomotion (reversals) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂, followed by a return to 7% O₂. (D) Locomotory speed of WT was recorded for 2 minutes at 7% O₂, followed by 148 minutes at 150 ppm H₂S balanced with 7% O₂, and then at 7% O₂. (E) Locomotory speed of WT animals in response to near-UV light (435 nm, 0.7 mW/mm²) exposure, before and directly after 30-minute preincubation in 150 ppm H₂S balanced with 7% O₂. Locomotory activity was recorded for 2 minutes before, during and after UV light exposure. (F) Locomotory speed of WT animals to different concentrations of H₂S balanced with 7% O₂. Red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively) used for statistical analysis. **** = $p < 0.0001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann-Whitney U test. (G) Distribution of WT animals in a microfluidic device after 25 minutes of aerotaxis. The gas inputs and the five chamber sections used for scoring are indicated. (H) Distribution of WT animals in a gradient of H₂S (150 ppm to 0 ppm). The bins correspond to different sections of the microfluidic chamber described in (G). N = 4 aerotaxis assays.

In the screen, a collection of mutants displayed reduced speed response, whereas only a few exhibited a lack of omega-turn and/or reversal responses to H₂S (**Supplementary file 1**). The reversal inhibition upon H₂S exposure and its rebound following H₂S withdrawal appeared to vary substantially across experiments for N2, suggesting that differences in population density, food availability, and animals' physiological state prior to the assay may matter. Nevertheless, our screen revealed that signaling pathways, including cGMP, insulin and TGF- β signaling, contributed to H₂S-evoked avoidance (**Supplementary file 1**). A naturally occurring variant of the neuropeptide receptor gene *npr-1*, previously implicated in the modulation of O₂ and CO₂ sensing, also affected locomotory speed in H₂S (**Supplementary file 1**). In addition, mitochondrial components, including those involved in respiration and H₂S detoxification, played critical roles in triggering and maintaining escape behavior in H₂S. In contrast, we did not find evidence that globins, K⁺ channels, or biogenic amine signaling contributed significantly to the behavioral avoidance to H₂S (**Supplementary file 1**).

cGMP signaling in ASJ cilia contributes to H₂S avoidance

One group of mutants exhibiting reduced locomotory speed in H₂S had defects in ciliogenesis (**Figure 2A–C**). This group included *daf-19* mutants lacking cilia, *dyl-3* mutants with reduced cilia length, and *dyl-7* mutants in which ciliogenesis occurs but the cilia are not anchored to their sensilla and are not exposed to the external environment (Heiman & Shaham, 2009; Murayama, Toh, Ohshima, & Koga, 2005; Starich et al., 1995; Swoboda, Adler, & Thomas, 2000). These findings suggest that ciliogenesis and exposure of cilia to the external environment are essential for animals to escape H₂S exposure. In addition, these mutants exhibited consistently high reversal rate, likely associated with their reduced locomotory speed (**Figure 2—figure supplement 1A–F**). Cilia-mediated sensory responses often involve the activation of guanylate cyclases and the opening of cGMP-gated channels (Ferkey, Sengupta, & L'Etoile, 2021). We observed significantly attenuated locomotory speed in response to H₂S in *tax-4* or *tax-2* mutants, which lack key subunits of cilia-enriched cGMP-gated channels (**Figure 2D**, **E**). In a comprehensive survey of all guanylate cyclase mutants, the *daf-11* strain emerged as the only mutant showing speed response defects in the avoidance of H₂S (**Figure 2D**, **supplementary file 1**). Similar to the cilia mutants, animals deficient in *daf-11* had low speed, attenuated omega turns, and constantly high reversal rate (**Figure 2D**, **Figure 2—figure supplement 1G, H**). Selective expression of *daf-11* in ASJ neurons, but not in other neurons, partially restored the speed response to acute H₂S (**Figures 2F**, **Figure 2—figure supplement 2A–D**). In addition, the H₂S response defect of *tax-4* mutants was also rescued by selective expression of its cDNA in ASJ neurons (**Figure 2G**), suggesting that the receptor guanylate cyclase DAF-11 and the CNG channel TAX-4 act at least partially in ASJ neurons to promote acute avoidance to H₂S. To validate the role of ASJ neurons in H₂S responses, we blocked neurotransmission from ASJ using the catalytic domain of *Tetanus Toxin* (TeTx), which impairs neurosecretion by specifically cleaving synaptobrevin (Schiavo et al., 1992). Cell-specific expression of TeTx in ASJ significantly inhibited the speed response to H₂S (**Figure 2H**). Surprisingly, using the Ca²⁺ sensor GCaMP6s to monitor a direct response of ASJ neurons to H₂S, we observed no H₂S-induced Ca²⁺ transients under any of the tested conditions, whereas CO₂-evoked Ca²⁺ increases were readily detected (**Figure 2—figure supplement 2E, F**). These findings suggest that although ASJ neuronal activity and neurosecretion contribute to the H₂S responses, ASJ neurons are unlikely to serve as the primary H₂S sensor.

Starvation modulates H₂S avoidance

In the candidate gene screen, another set of mutants with reduced speed responses to H₂S had previously been shown to be defective in response to CO₂ (Hallem & Sternberg, 2008). Specifically, the H₂S-evoked speed response was reduced in mutants deficient in nutrient-sensitive signaling, including the insulin receptor DAF-2, the TGF- β ligand DAF-7, and the nuclear hormone receptor NHR-49 (**Figure 3A–C**, **Figure 3—figure supplement 1A**). Similarly, mutants lacking other CO₂ response modulators, such as the calcineurin subunits TAX-6 and CNB-1, also failed to

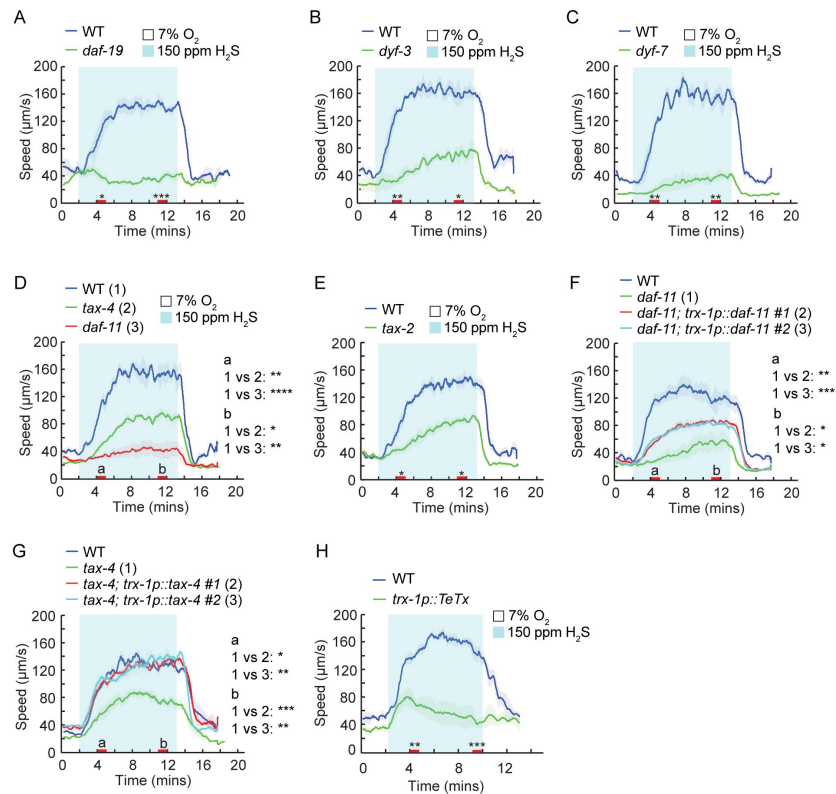


Figure 2.

Neurosecretion from ASJ neurons contributes to H₂S avoidance

(A–H) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and *daf-19(m86)* (A); WT and *dyf-3(m185)* (B); WT and *dyf-7(m539)* (C); WT, *daf-11(m47)* and *tax-4(p678)* (D); WT and *tax-2(p691)* (E); WT, *daf-11(m47)* and transgenic *daf-11(m47)* expressing *daf-11* genomic DNA in ASJ neurons (two independent lines) (F); WT, *tax-4(p678)* and transgenic *tax-4(p678)* expressing *tax-4* cDNA in ASJ neurons (two independent lines) (G); WT and transgenic WT expressing the catalytic domain of the *tetanus* toxin (TeTx) in ASJ neurons (H). For the comparison of acute locomotory speed responses between strains, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively, in panels D, F, and G) used for statistical analysis. **** = $p < 0.0001$, *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann-Whitney U test.

respond to H₂S (**Figure 3—figure supplement 1B, C**). In addition to the abolished speed response, we also observed that the H₂S-evoked omega-turn response was nearly abolished in *daf-2* mutants, and high proportion of animals exhibited persistently high reversal rate regardless of H₂S stimulation (**Figure 3—figure supplement 1D–F**). The H₂S avoidance defects observed in *daf-2* and *daf-7* single mutants were fully suppressed in *daf-2; daf-16* and *daf-7; daf-3* double mutants (**Figure 3A, B**). Therefore, it is likely that insulin and TGF- β signaling modulate H₂S responses by regulating the expression of relevant genes via DAF-16 and DAF-3 transcription factors, respectively. Given that H₂S-evoked speed response was modulated by the nutrient-sensitive pathways, we further explored whether the locomotory response to H₂S was sensitive to the nutrient state. Similar to the response to CO₂ (Bretscher et al., 2008; Hallem & Sternberg, 2008), we observed that a 24-hour period of starvation suppressed the speed response to H₂S (**Figure 3D**). However, the CO₂ sensor GCY-9 was dispensable for H₂S-evoked avoidance (**Figure 3—figure supplement 1G–I**). These findings suggest that even though acute avoidance of CO₂ and H₂S are modulated through shared mechanisms, distinct molecular sensors are used.

The O₂ sensing circuit antagonizes H₂S avoidance in wild isolates

Wild strains of *C. elegans* thrive in environmental niches characterized by variable O₂, CO₂ and H₂S levels (Adams et al., 1979; Bretscher et al., 2008; Budde & Roth, 2011; Gea Barrena, Artola, & Sanchez, 2004; Hallem & Sternberg, 2008; Morra & Dick, 1991; Oshins, Michel, Louis, Richard, & Rynk, 2022; Patange et al., 2025; Rodríguez-Kabana et al., 1965). Compared to the laboratory reference strain N2, wild isolates display robust responses to high O₂ but reduced sensitivity to CO₂ stimulation (Beets et al., 2020; Bretscher et al., 2008; Carrillo, Guillermin, Rengarajan, Okubo, & Hallem, 2013; Hallem & Sternberg, 2008; Kodama-Namba et al., 2013; McGrath et al., 2009). It is thought that the unique O₂ and CO₂ responses of N2 strain evolved during its domestication on solid media at atmospheric gas concentrations (Sterken, Snoek, Kammenga, & Andersen, 2015). We hypothesized that wild *C. elegans* strains might also exhibit an attenuated response to H₂S, and therefore included a set of wild strains in our candidate gene screen. Consistent with this hypothesis, wild isolates including CB4855, CB4856, and CB4858 exhibited reduced locomotory speed in H₂S (**Figure 3E**, **Figure 3—figure supplement 1j**).

A naturally occurring variation in the neuropeptide receptor NPR-1 predominantly determines the differences in locomotory responses to O₂ and CO₂ between the laboratory N2 strain and wild isolates. The NPR-1(215F) variant found in wild isolates is less active than the NPR-1(215V) found in N2 animals (de Bono & Bargmann, 1998). We wondered whether the variations in the *npr-1* gene also contribute to differences in the acute response to H₂S. The absence of *npr-1* was sufficient to reduce the speed response to H₂S, suggesting that active NPR-1 signaling promotes H₂S avoidance (**Figure 3F**). NPR-1(215V) primarily acts within the RMG interneurons to modulate responses to environmental stimuli by inhibiting signal output from these neurons (**Figure 3G**) (Laurent et al., 2015; Macosko et al., 2009). Cell-specific expression of the highly active *npr-1*(215V) variant in RMG neurons effectively restored the H₂S-evoked locomotory speed in *npr-1* null mutant worms (**Figure 3H**). Furthermore, optogenetic stimulation of RMG interneurons using channelrhodopsin-2 (ChR2) significantly dampened the locomotory response to H₂S in N2 animals, suggesting that high RMG activity inhibits H₂S responses (**Figure 3—figure supplement 1K**).

The perception of 21% O₂ involves the soluble guanylate cyclases GCY-35/GCY-36 and the downstream cyclic nucleotide-gated (CNG) channels TAX-4/TAX-2 in the URX, AQR, and PQR O₂-sensing neurons, which relay the signals to RMG (Busch et al., 2012; Cheung, Cohen, Rogers, Albayram, & de Bono, 2005; Couto, Oda, Nikolaev, Soltesz, & de Bono, 2013; Gray et al., 2004; Laurent et al., 2015; Persson et al., 2009; Zimmer et al., 2009). We investigated whether disrupting the O₂ sensing machinery would have the same effect as inhibiting RMG signaling pathway by NPR-1(215V). Mutating *gcy-35* and *tax-4*, or genetically ablating the O₂ sensing neurons URX, AQR, and PQR by cell-specific expression of the pro-apoptotic gene *egl-1*

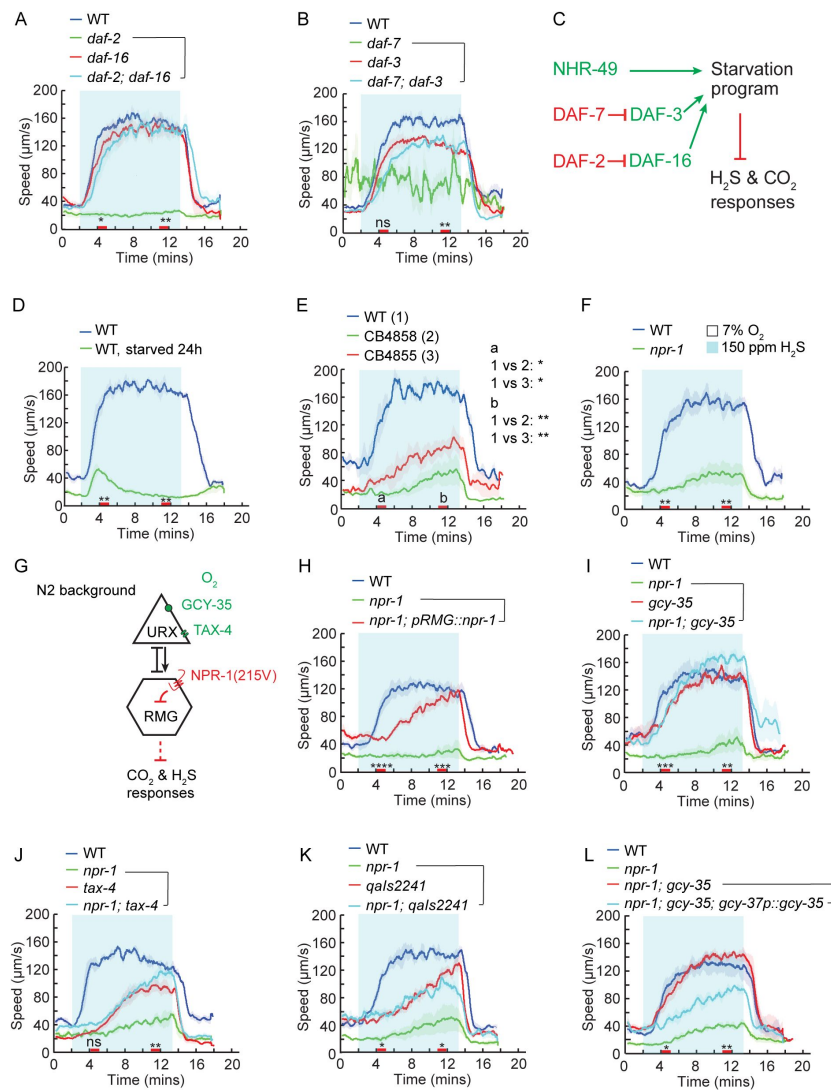


Figure 3.

Insulin, TGF- β , and high O₂ signaling pathways antagonizes H₂S avoidance

(A, B) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *daf-2(e1370)*, *daf-16(mgDf47)*, and *daf-2(e1370); daf-16(mgDf47)* double mutants (A); WT, *daf-7(e1372)*, *daf-3(mgDf90)*, and *daf-7(e1372); daf-3(mgDf90)* double mutants (B). For the *daf-2* assays, WT and *daf-2* mutants were maintained at 15°C. L4 animals were picked and shifted to 25°C until day-one adults, then assayed at room temperature. (C) Hypothetical model for the regulation of H₂S avoidance by a starvation program involving the insulin, TGF- β , and NHR-49 pathways. (D) Locomotory speed response to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ in fed and starved WT animals. (E, F) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and wild isolates (CB4858 and CB4855) (E); WT and *npr-1(ad609)* (F). (G) Hypothetical model showing how O₂ regulates H₂S avoidance via the inhibitory RMG circuit activity, which is modulated by O₂ sensory inputs and NPR-1 signaling. (H-L) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *npr-1(ad609)*, and transgenic *npr-1(ad609)* expressing *npr-1* in RMG neurons using Cre-LoxP system (Macosko et al., 2009) (H); WT, *npr-1(ad609)*, *gcy-35(ok769)*, and *npr-1(ad609); gcy-35(ok769)* double mutants (I); WT, *npr-1(ad609)*, *tax-4(p678)* and *npr-1(ad609); tax-4(p678)* double mutants (J); WT, *npr-1(ad609)*, *qals2241* (genetic ablation of AQR, PQR and URX neurons) and *npr-1(ad609); qals2241* (K); WT, *npr-1(ad609)*, *npr-1(ad609); gcy-35(ok769)* double mutants and transgenic *npr-1(ad609); gcy-35(ok769)* double mutants expressing *gcy-35* cDNA under *gcy-37* promoter, which drives *gcy-35* expression in O₂ sensing neurons (L). For the comparison of acute locomotory responses between strains, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively, in panel E) used for statistical analysis. **** = $p < 0.0001$, *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann-Whitney U test.

restored H₂S-evoked speed response in *npr-1* null mutants (**Figure 3I–K**). Specific expression of *gcy-35* cDNA in O₂ sensing neurons was sufficient to repress this response to H₂S in *npr-1*; *gcy-35* double mutants (**Figure 3L**). Finally, enhancing the presynaptic activity of URX, AQR, PQR neurons by the expression of a gain-of-function PKC-1(A160E) (Dekker, McIntyre, & Parker, 1993; Hiroki et al., 2022) significantly dampened H₂S-induced locomotory speed (**Figure 3—figure supplement 1L**). Taken together, these observations indicate that activation of the O₂-sensing circuit inhibits the H₂S-evoked avoidance response.

H₂S exposure reprograms gene expression in *C. elegans*

Wild isolates that were relatively insensitive to H₂S exposure (**Figure 3E**, **Figure 3—figure supplement 1J**) likely developed sensory and physiological adaptations to persist in environments where H₂S may transiently increase (Budde & Roth, 2011; Patange et al., 2025; Rodriguez-Kabana et al., 1965; Romanelli-Cedrez et al., 2024). These adaptations to H₂S are mediated, at least in part, by the reprogramming of gene expression (Miller et al., 2011a). To identify the genes whose expression was induced by prolonged H₂S exposure, we conducted a comparative analysis of transcriptome profiles in animals exposed to 50 ppm or 150 ppm H₂S for 1 hour, 2 hours, or 12 hours. RNA-seq analysis revealed that exposure to either 50 ppm or 150 ppm H₂S for one hour was sufficient to trigger significant changes in gene expression, with 518 or 304 genes showing differential expression, respectively (**Figure 4A**, **Supplementary file 2**). Genes induced by H₂S exposure for 1 or 2 hours displayed considerable overlap (**Figure 4B**, **Figure 4—figure supplement 1A**). Gene Ontology (GO) analysis revealed that biological processes such as defense against bacteria and cysteine biosynthesis from L-serine were significantly enriched after H₂S exposure (**Figure 4C, D**, **Figure 4—figure supplement 1B, C**). As expected, we observed a robust induction of genes involved in H₂S detoxification (**Figure 4E, F**) (Miller, Budde, & Roth, 2011b; Niu et al., 2011; Vora et al., 2022). H₂S is detoxified in mitochondria by sulfide:quinone oxidoreductase SQRD-1 to produce persulfide, which is then metabolized by sulfur dioxygenase ETHE-1 to generate sulfite. Thiosulfate transferase further oxidizes sulfite to produce thiosulfate (Hildebrandt & Grieshaber, 2008). Glutathione S-transferases (GSTs) remove accumulated sulfur molecules during H₂S oxidation (Jackson, Melideo, & Jorns, 2012). Notably, *gst-19* and *sqr-1* were among the most significantly upregulated genes after one or two hours of exposure to either 50 ppm or 150 ppm H₂S, while *ethe-1* showed a weak increase (**Figure 4E, F**, **Figure 4—figure supplement 1D, E**, **Supplementary file 2**).

H₂S exposure is known to activate the HIF-1 pathway (Budde & Roth, 2010, 2011; Ma, Vozdek, Bhatla, & Horvitz, 2012; Miller et al., 2011a; Miller & Roth, 2007; Vora et al., 2022). In line with this, a proportion of genes with HIF-1 regulated promoters displayed increased expression after 1 or 2 hours of H₂S exposure (**Figure 4—figure supplement 1F**). Among these were the aforementioned detoxifying enzymes GST-19, SQRD-1, and ETHE-1, as well as CYSL-1, CYSL-2, and CDO-1, which are involved in cysteine metabolism (**Supplementary file 2**). Consistent with earlier studies, we also observed differential expression of a set of SKN-1 targets in response to chronic H₂S exposure (**Figure 4—figure supplement 1G**) (Horsman et al., 2019; Miller et al., 2011a; Niu et al., 2011). This included the heat shock protein genes *hsp-16.2* and *hsp-16.41* (**Figure 4E, F**, **Figure 4—figure supplement 1D, E**). Surprisingly, we detected a rapid increase in the expression of genes associated with intracellular H₂S production, including the methanethiol oxidase-encoding gene *semo-1* and the mercaptopyruvate sulfurtransferase-encoding gene *mpst-3* (**Figure 4E, F**, **Figure 4—figure supplement 1D, E**) (Philipp et al., 2022; Qabazard et al., 2014). Elevated expression of 3-mercaptopyrivate sulfurtransferase has similarly been reported in sulfide spring fish, suggesting a conserved response mechanism to H₂S (Kelley et al., 2016; Mathew, Schlupalius, & Ebert, 2011; Qabazard et al., 2014). However, the functional implications of increased expression of H₂S synthesis genes remains unclear.

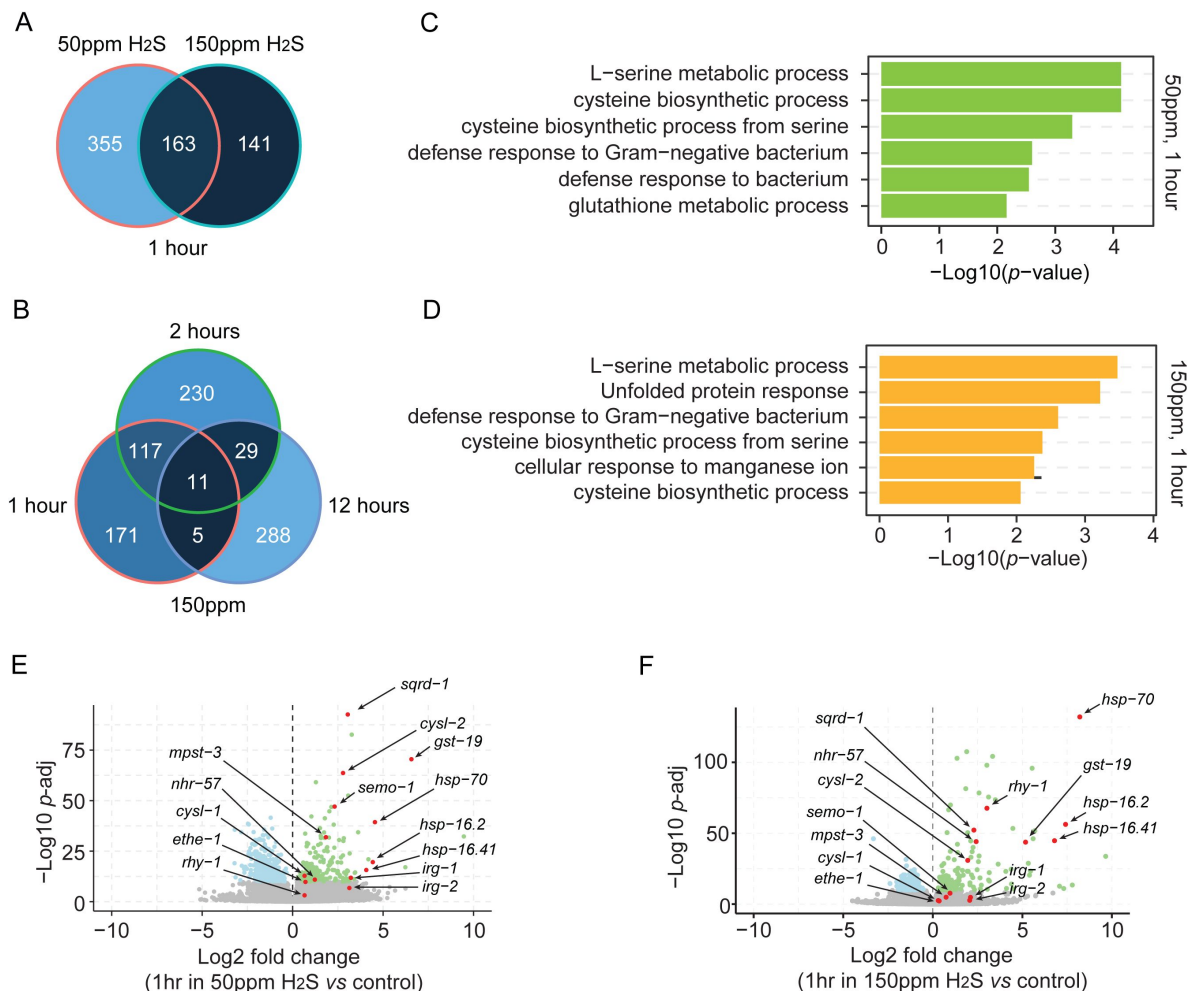


Figure 4.

Prolonged H₂S exposure reprograms gene expression in *C. elegans*

(A) Venn diagram displaying the number of differentially expressed genes after 1-hour exposure to 50 ppm or 150 ppm H₂S balanced with 7% O₂ in WT animals (adjusted $p < 1e-10$). (B) Venn diagram displaying the number of differentially expressed genes after 1-, 2-, and 12-hour exposure to 150 ppm H₂S balanced with 7% O₂ in WT animals (adjusted $p < 1e-10$). (C, D) Significantly enriched GO categories for differentially expressed genes with adjusted $p < 1e-10$ in WT animals exposed to 50 ppm (C) or 150 ppm (D) H₂S balanced with 7% O₂ for 1 hour. (E, F) Volcano plots showing the differentially expressed genes with adjusted $p < 1e-10$ in WT animals exposed to 50 ppm (E) or 150 ppm (F) H₂S balanced with 7% O₂ for 1 hour. A set of genes involved in H₂S detoxification, cysteine metabolism and stress response were highlighted in red.

When exposed to H₂S for 12 hours, the differentially expressed genes showed reduced overlap with those identified at earlier time points (**Figure 4B**, **Figure 4—figure supplement 1A**), suggesting that distinct defense strategies may be employed at various stages of H₂S exposure. Furthermore, substantial overlap was observed between the sets of genes induced by 50 ppm and 150 ppm H₂S after 1h or 2h exposure. However, this overlap decreased significantly after 12 hours exposure (**Figure 4—figure supplement 1H–J**). Additionally, we noticed that 150 ppm H₂S specifically triggered the expression of the HIF-1 regulator *rhy-1* and the HIF-1 target *nhr-57* at all exposure time points, a response that was less pronounced at 50 ppm H₂S (**Figures 4E, F**, **Figure 4—figure supplement 1D, E, K** and **L**). Together, these findings suggest that multiple defense mechanisms are induced to adapt to the stress imposed by H₂S, which can diverge over time and at different H₂S concentrations.

Locomotory speed response to H₂S is

modulated by HIF-1-induced detoxification

Given that HIF-1 signaling is robustly activated shortly after H₂S exposure (**Figure 4—figure supplement 1F**) (Budde & Roth, 2010; 2011; Miller et al., 2011a), we next sought to determine how HIF-1 signaling might contribute to H₂S-evoked avoidance response, and began by examining the effects of HIF-1 stabilization. Acclimating animals in 1% O₂ for 12 hours or longer significantly decreased the speed response to H₂S, indicating that HIF-1 stabilization inhibits the H₂S-evoked avoidance behavior (**Figure 5A**). To confirm this observation, we examined several mutants with stabilized HIF-1. The conserved proline-4-hydroxylase PHD/EGL-9 and the von Hippel-Lindau (VHL) tumor suppressor protein are required for HIF-1 degradation (Kaelin & Ratcliffe, 2008; Semenza, 2010). Disruption of either *egl-9* or *vhl-1* leads to HIF-1 stabilization under normoxic conditions. Similar to the effect of prolonged hypoxia, both mutants exhibited a markedly reduced locomotory speed in H₂S (**Figure 5B**). Interestingly, the omega-turn and reversal responses to H₂S were also substantially inhibited in *egl-9* mutants (**Figure 5—figure supplement 1A, B**). In contrast, the locomotory speed response to 1% O₂ remained largely unaffected in *egl-9* deficient animals (**Figure 5—figure supplement 1C**), suggesting a selective impairment of the H₂S response. In addition, expressing the non-degradable forms of HIF-1, *hif-1*(P621A or P621G), under the pan-neuronal *unc-14* promoter or under the endogenous *hif-1* promoter, was sufficient to inhibit H₂S evoked speed responses (**Figure 5C**). Together, these findings indicate that the activation of HIF-1 signaling during prolonged H₂S or hypoxia exposure mediates an adaptive response associated with reduced behavioral responses to H₂S, and that HIF-1 stabilization in neurons alone is sufficient to promote this behavioral adaptation.

On the other hand, animals lacking HIF-1 signaling have previously been shown to display increased sensitivity to H₂S and become paralyzed rapidly upon exposure (Budde & Roth, 2010; Horsman et al., 2019). Consistent with this, *hif-1* mutants exhibited a brief increase in speed upon H₂S exposure, which rapidly declined to baseline levels (**Figure 5D**). They also displayed a stronger initial omega-turn response and greater inhibition of the reversal rate than WT (**Figure 5E, F**), confirming that animals deficient in HIF-1 signaling are sensitized to H₂S exposure. This transient avoidance behavior was also observed in *egl-9*; *hif-1* and *vhl-1*; *hif-1* double mutants (**Figure 5G, H**), suggesting that this phenotype is linked to the absence of HIF-1 signaling. Importantly, *hif-1* mutants displayed a robust speed response to other stimuli, such as near-UV light and 1% O₂ stimulation (**Figure 5I**, **Figure 5—figure supplement 1D**), suggesting that the brief avoidance response is specific to H₂S rather than reflecting a general locomotory defect. Further supporting increased sensitivity to H₂S in *hif-1* mutants, the locomotory speed response of *hif-1* mutants to subsequent near-UV light stimulation was nearly abolished after 30 minutes of H₂S exposure, whereas *egl-9* mutants and WT animals remained fully responsive (**Figure 5I**). The reduced movement and lack of responsiveness to other stressors in *hif-1* mutants resemble the stress-induced sleep-like behavior previously observed in response to noxious heat (Byrne Rodgers & Ryu, 2020). Prompted by this observation, we next explored whether HIF-1-induced detoxification genes were also required to maintain H₂S-evoked behavioral responses (**Figure 5—**

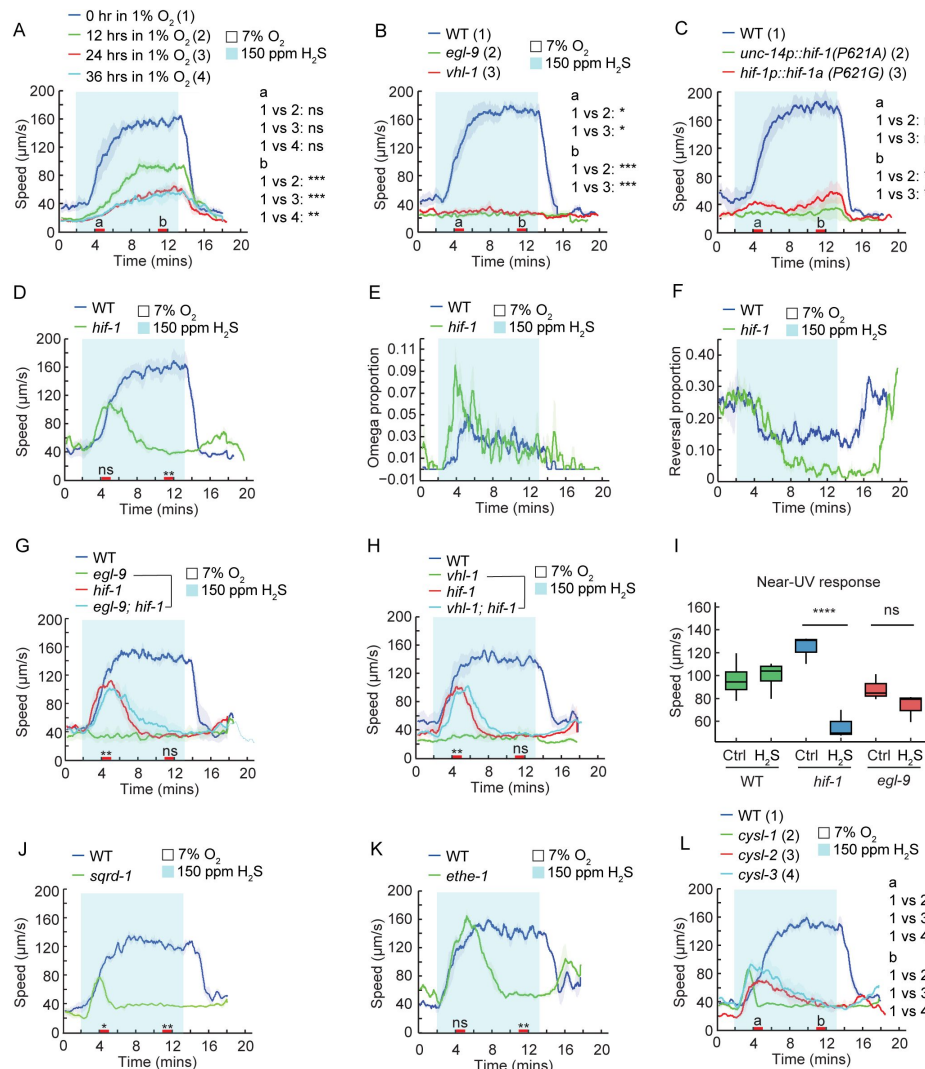


Figure 5.

Acute response to H₂S is modulated by HIF-1 signaling

(A) Locomotory speed responses of WT animals to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ after incubation of the animals for 0, 12, 24, or 36 hours in 1% O₂. (B, C) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *egl-9(sa307)*, and *vh1-1(ok161)* (B); WT and transgenic animals expressing the non-degradable form of HIF-1 (P621A or P621G) under the pan-neuronal *unc-14* promoter or the endogenous *hif-1* promoter, respectively (C). (D–F) Avoidance responses of WT and *hif-1(ia4)* to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂, including locomotory speed (D), reorientation (omega-turn) (E), and reversal (F). (G, H) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *egl-9(sa307)*, *hif-1(ia4)*, and *egl-9(sa307); hif-1(ia4)* double mutants (G); WT, *vh1-1(ok161)*, *hif-1(ia4)*, and *vh1-1(ok161); hif-1(ia4)* double mutants (H). (I) Average locomotory speed during 2 minutes of exposure to near-UV light for WT, *hif-1(ia4)*, and *egl-9(sa307)* before (Ctrl = control) and after 30 minutes of exposure to 150 ppm H₂S balanced with 7% O₂. (J–L) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and *sqrd-1(tm3378)* (J); WT and *ethe-1(yum2895)* (K); WT, *cysl-1(ok762)*, *cysl-2(ok3516)*, and *cysl-3(yum4)* (L). For the comparison of acute locomotory responses between strains, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively, in panels A, B, C, and L) used for statistical analysis. *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann-Whitney U test.

figure supplement 1E [↗](#)). Similar to *hif-1* mutants, *sqr-1* and *ethe-1* mutants displayed brief responses to H₂S, with locomotory activity rapidly returning to baseline levels (**Figure 5J, K** [↗](#), **Figure 5—figure supplement 1F–I** [↗](#)). However, both mutants responded normally to 1% O₂, suggesting a specific hypersensitivity to H₂S toxicity (**Figure 5—figure supplement 1J** [↗](#)). Furthermore, we examined the contribution of additional genes whose expression was upregulated in response to H₂S, including the sulfhydrylase/cysteine synthase-encoding genes *cysl-1*, *cysl-2*, and *cysl-3*, as well as *semo-1*. Disrupting any of these genes phenocopied the responses observed in *hif-1*, *ethe-1*, or *sqr-1* mutants, while they displayed normal responses to 1% O₂ (**Figure 5L** [↗](#), **Figure 5—figure supplement 1D, K–P** [↗](#)). These observations suggest that while the HIF-1-induced H₂S detoxification system is not essential for the initial sensing of H₂S, it is required to maintain animals' locomotory activity in the presence of H₂S.

Labile iron pool sustains the locomotory activity in H₂S

Among the most differentially regulated genes upon exposure to H₂S, we observed consistent downregulation of *ftn-1* and to a lesser extent *ftn-2*, whereas *smf-3* expression was increased under specific H₂S conditions (**Figure 6A, B** [↗](#), **Figure 6—figure supplement 1A–D** [↗](#)). The ferritin-encoding genes *ftn-1* and *ftn-2*, along with the iron transporter *smf-3*, are critical for maintaining intracellular labile iron homeostasis in *C. elegans* (Ackerman & Gems, 2012 [↗](#); Gourley, Parker, Jones, Zumbrennen, & Leibold, 2003 [↗](#); Rajan et al., 2019 [↗](#); Romney, Newman, Thacker, & Leibold, 2011 [↗](#)), suggesting that H₂S exposure may interfere with cellular iron storage. It has been shown that disrupting *ftn-1* increases labile iron pools in the cytoplasm, while its overexpression reduces these levels (Anderson & Leibold, 2014 [↗](#); Romney et al., 2011 [↗](#)). We found that *ftn-1* mutants had enhanced speed response to H₂S, whereas *ftn-1* overexpression attenuated H₂S-induced speed increase without damaging acute response to 1% O₂ (**Figure 6C, D** [↗](#), **Figure 6—figure supplement 1E** [↗](#)). In addition, *smf-3* mutants, which impair iron uptake, exhibited a reduced speed response to H₂S (**Figure 6D** [↗](#)). These results support the idea that H₂S exposure disrupts iron homeostasis.

To complement our genetic analysis, we pharmacologically depleted labile irons. In the presence of the iron chelator 2,2'-dipyridyl (BP), the H₂S-evoked speed response was fully lost (**Figure 6E** [↗](#)), whereas locomotory speed in response to hypoxia was maintained (**Figure 6—figure supplement 1F** [↗](#)). We noticed that, similar to *hif-1* mutants, both *smf-3* mutants and BP-treated animals exhibited a strong initial omega-turn response to H₂S (**Figure 6—figure supplement 1G, I** [↗](#)), suggesting that these animals are hypersensitive to H₂S toxicity. One possibility is that low iron levels compromise H₂S detoxification capacity, as the detoxification enzyme ETHE-1 requires iron for its activity (Kabil & Banerjee, 2012 [↗](#); Pettinati, Brem, McDonough, & Schofield, 2015 [↗](#)). By contrast, increasing iron levels by feeding ferric ammonium citrate (FAC) or by disrupting *ftn-1* significantly sustained animals' locomotory speed in H₂S (**Figure 6D, E** [↗](#)), likely by delaying H₂S-mediated iron depletion. However, the reorientation and reversals were not significantly affected (**Figure 6—figure supplement 1G–J** [↗](#)). This impact of FAC supplementation and *ftn-1* disruption on speed response was largely lost in the absence of *hif-1* (**Figure 6F, G** [↗](#)), suggesting that the effects of iron require HIF-1-dependent detoxification (**Figure 6H** [↗](#)). Yet, increased iron availability modestly improved the locomotory activity of *hif-1* mutants in H₂S (**Figure 6F, G** [↗](#)), presumably by compensating for the low iron levels in *hif-1* mutant animals (Rajan et al., 2019 [↗](#); Romney et al., 2011 [↗](#)).

Mitochondrial function modulates H₂S-evoked locomotion responses

The mitochondrial electron transport chain (ETC) is not only the primary target of H₂S toxicity, but is also an essential component of the H₂S detoxification pathways (Cooper & Brown, 2008 [↗](#); Khan et al., 1990 [↗](#); Nicholls & Kim, 1982 [↗](#); Romanelli-Cedrez et al., 2024 [↗](#)). Based on this, we hypothesized that ETC plays a central role in the locomotory response to H₂S. Indeed, mutating the genes critical for ETC function, including *gas-1*, *clk-1*, *mev-1*, and *isp-1*, impaired the H₂S-evoked

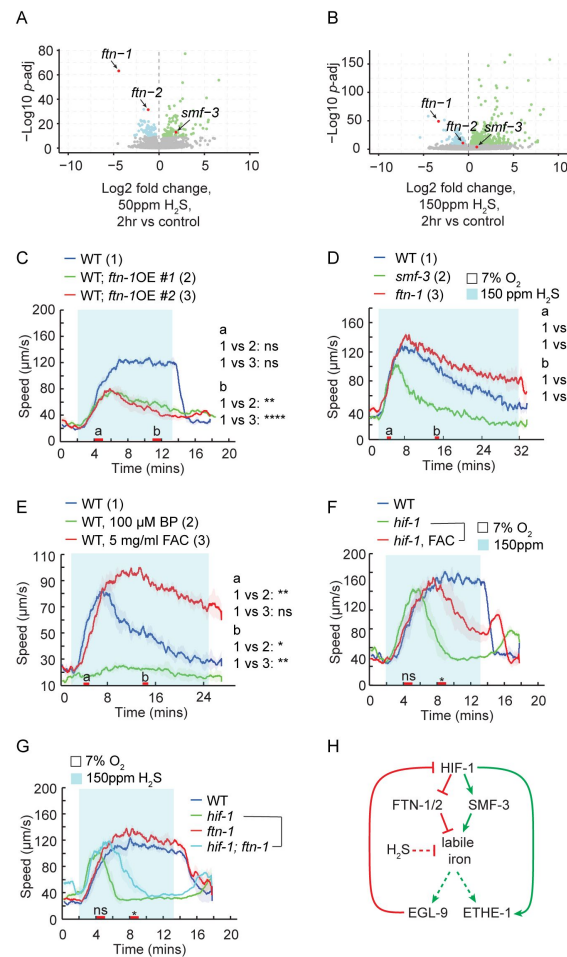


Figure 6.

Labile iron pool sustains the locomotory activity in H₂S

(A, B) Volcano plots showing the relative expression of the genes involved in the regulation of iron homeostasis in WT animals after 2-hour exposure in 50 ppm (A) or 150 ppm (B) H₂S balanced with 7% O₂. (C–G) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of indicated genotypes or treatments: WT and animals overexpressing *ftn-1* genomic DNA under its own promoter (#1 and #2 indicate two independent lines) (C); WT, *smf-3(ok1305)*, and *ftn-1(ok3625)* (D); WT, and WT pretreated with 100 μM 2,2'-Bipyridyl (BP) or with 5 mg/ml ferric ammonium citrate (FAC) in the presence of food for 16 hours (E); WT, *hif-1(ia4)*, and *hif-1(ia4)* mutants pretreated with 5 mg/ml ferric ammonium citrate (FAC) for 16 hours (F); WT, *hif-1(ia4)*, *ftn-1(ok3625)*, and *hif-1(ia4); ftn-1(ok3625)* double mutants (G). (H) Hypothetical model of the interactions between labile iron pool and HIF-1 signaling. For the comparison of acute locomotory speed responses between strains or conditions, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively, in panels C, D, and E) used for statistical analysis. **** = $p < 0.0001$, *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann–Whitney U test.

speed and reversal responses (**Figure 7A–D**, **Figure 7—figure supplement 1A–D**). These data suggest that a functional ETC is required to trigger and support H₂S-evoked avoidance behavior or that dysfunctional ETC induces an adaptive response to cellular stressors. Consistently, exposing animals to rotenone, an ETC complex I inhibitor, robustly inhibited the speed responses to H₂S at all time points we tested (**Figure 7E**). Interestingly, we noticed that transient rotenone exposure substantially increased animals' basal locomotory speed (**Figure 7E**) (Onukwufo et al., 2022), suggesting that acute ETC interruption evokes behavioral avoidance, in a manner similar to that observed with other noxious stimuli. However, a 2-hour exposure to rotenone abolished the H₂S-evoked speed response without increasing the basal locomotory speed (**Figure 7E**), suggesting that prolonged ETC inhibition may either impair the cellular capacity to support H₂S-evoked behavioral responses or drive animals' adaptation to cellular stress and consequently reduce their locomotory response to H₂S.

Given that rotenone exposure and mutations in ETC genes increase mitochondrial ROS generation (Ishii et al., 2013; Ochi et al., 2016; Ramsay & Singer, 1992; Yang & Hekimi, 2010; Zorov, Juhaszova, & Sollott, 2014), and that high levels of H₂S also induce ROS production (Jia et al., 2020), we sought to determine how mitochondrial ROS contributes to speed response to high H₂S. We focused on superoxide dismutases (SODs), key enzymes in superoxide detoxification. The *C. elegans* genome encodes five *sod* genes: *sod-1* and *sod-5* encode cytosolic Cu/ZnSODs, *sod-2* and *sod-3* encode mitochondrial MnSODs, and *sod-4* encodes the extracellular Cu/ZnSOD isoforms (Doonan et al., 2008; Zubovych, Straud, & Roth, 2010). All *sod* single mutants exhibited a robust initial response to 150 ppm H₂S (**Figure 7F, G**). Interestingly, the mitochondrial SOD mutants *sod-2* and *sod-3*, as well as *sod-5*, showed an initial burst of locomotory activity followed by a rapid decline in speed in the presence of H₂S (**Figure 7F, G**), but maintained a high locomotory speed in response to 1% O₂ (**Figure 7—figure supplement 1E**). In the quintuple *sod-1; sod-2; sod-3; sod-4; sod-5* mutant, which lacks all superoxide dismutases and presumably accumulates higher ROS levels, the H₂S-evoked speed and omega-turn responses were nearly abolished, and the proportion of animals exhibiting reversals remained consistently high, a pattern resembling that of *egl-9* and ETC mutants (**Figure 7G**, **Figure 7—figure supplement 1F, G**). Similar to *sod-2* and *sod-3*, the quintuple *sod* mutant exhibited robust speed response to 1% O₂ (Zhao et al., 2022). In addition, the quintuple mutants responded normally to near-UV light after 30 minutes of H₂S exposure (**Figure 7—figure supplement 1H**), indicating that their neuromuscular function remains largely preserved. Therefore, high ROS levels in *sod* mutants do not substantially compromise the detoxification capacity required for survival in H₂S. Instead, high ROS appears to promote adaptation to H₂S-induced stress, presumably by activating several stress-responsive pathways such as HIF-1, NRF2/SKN-1, and DAF-16/FOXO (**Figure 7H**) (Lennicke & Cocheme, 2021; Patten, Germain, Kelly, & Slack, 2010).

Discussion

Animals' behavior and physiology are profoundly influenced by the environments in which they evolved. The laboratory strain N2, which is adapted to low atmospheric CO₂ and H₂S concentrations, robustly avoids both gases (Beets et al., 2020; Bretscher et al., 2008; Carrillo et al., 2013; Hallem & Sternberg, 2008; Kodama-Namba et al., 2013; McGrath et al., 2009). Upon acute exposure to H₂S above 75 ppm, N2 exhibits an avoidance behavior characterized by reorientation and increased locomotion. This response is significantly reduced in wild isolates. These divergent responses between the laboratory strain and wild isolates are primarily driven by variations in the *npr-1* gene, which encodes a neuropeptide receptor, suggesting that *C. elegans* has undergone rapid evolutionary adaptation to its environment.

The behavioral response to a specific sensory stimulus is usually shaped by an interplay of multiple environmental and physiological cues. We observe that conditions dampening *C. elegans*' response to CO₂ also impair the response to H₂S. Specifically, the speed response to H₂S is

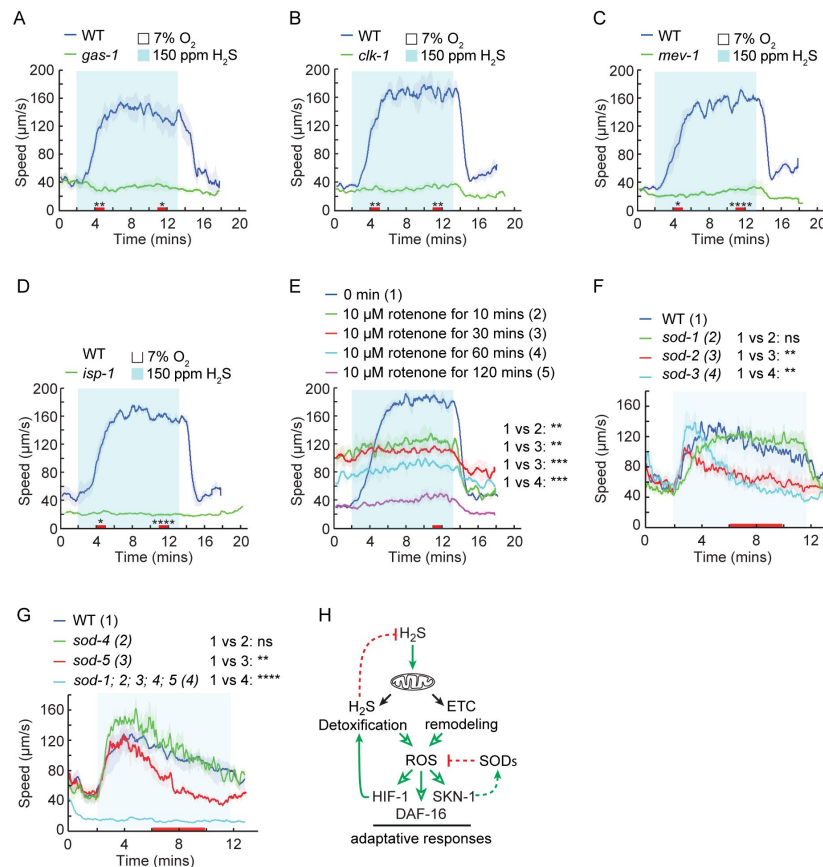


Figure 7.

Mitochondrial function is required for acute response to H₂S

(A–D) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and *gas-1(fc21)* (A); WT and *clk-1(qm30)* (B); WT and *mev-1(kn1)* (C); WT and *isp-1(qm150)* (D). (E) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT animals pretreated with 10 μM rotenone for 0 minute, 10 minutes, 30 minutes, 60 minutes or 120 minutes. For the comparison of acute locomotory speed responses between strains or conditions, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes) used for statistical analysis. *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann–Whitney U test. (F–G) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *sod-1(tm776)*, *sod-2(ok1030)*, and *sod-3(tm760)* (F); WT, *sod-4(gk101)*, *sod-5(tm1146)*, and *sod-1(tm783)*; *sod-2(ok1030)*; *sod-3(tm760)*; *sod-4(gk101)*; *sod-5(tm1146)* (G). For the comparison of acute locomotory speed responses between strains or conditions, red bars on the x-axis represent the time interval (6–10 minutes) used for statistical analysis. *** = $p < 0.001$, ** = $p < 0.01$, * = $p < 0.05$, ns = not significant, Mann–Whitney U test. (H) Hypothetical model of the role of mitochondria in response to toxic levels of H₂S. In this model, mitochondria play a dual role in H₂S-evoked avoidance behavior. A transient burst of mitochondrial ROS triggered by high H₂S levels initiates locomotory avoidance, whereas sustained ROS elevation activates stress-responsive pathways, including HIF-1, NRF2/SKN-1, and DAF-16/FOXO, promoting adaptation to prolonged H₂S exposure.

significantly inhibited at high O₂ levels, and in mutants with defective insulin or TGF- β pathway that signal starvation. However, despite these shared modulation, CO₂ and H₂S sensing involve distinct molecular mechanisms. For example, CO₂ responses are mediated by the guanylate cyclase GCY-9 in BAG neurons (Hallem et al., 2011 [\[1\]](#)), which is dispensable for H₂S sensing. In addition, acute exposure to H₂S induces stronger reorientation but delayed acceleration compared to CO₂, suggesting different sensing and information processing mechanisms (Bretscher et al., 2008 [\[2\]](#)). Therefore, even though responses to CO₂ and H₂S are modulated by overlapping regulatory mechanisms, the two gases are detected by distinct sensors and processed differently.

In a candidate gene survey, we excluded the potential involvement of globins, K⁺ channels, biogenic amines, and most guanylate cyclases in H₂S responses. However, we observed that H₂S avoidance requires the activity of guanylate cyclase DAF-11 in ASJ neurons, as well as neurosecretion from these neurons. Since the DAF-11 pathway and neurosecretion from ASJ neurons regulate developmental programs that modify sensory functions in *C. elegans* (Murakami, Koga, & Ohshima, 2001 [\[3\]](#)), it is not surprising that *daf-11* mutants display pleiotropic phenotypes including impaired H₂S and CO₂ responses (Hallem & Sternberg, 2008 [\[4\]](#)). In addition, we did not detect an H₂S-evoked calcium transient in ASJ neurons. Therefore, although ASJ neurons and DAF-11 activity are clearly required, the exact sensory neurons, circuits, and molecular triggers mediating acute H₂S avoidance remain to be elucidated.

Consistent with previous reports (Horsman et al., 2019 [\[5\]](#); Miller et al., 2011a [\[6\]](#)), H₂S exposure substantially alters the gene expression in *C. elegans*, including those involved in H₂S detoxification and iron homeostasis pathways. The repression of *ftn-1* and induction of *smf-3* observed during H₂S exposure suggest that H₂S depletes intracellular labile iron. Iron is a critical cofactor required for the activity of many enzymes and supports a wide range of cellular processes (Dev & Babitt, 2017 [\[7\]](#); Hentze, Muckenthaler, & Andrews, 2004 [\[8\]](#)). One important enzyme that requires labile iron as cofactor is PHD/EGL-9, which targets HIF-1 for degradation. Changes in iron availability are therefore expected to modulate EGL-9 activity, which in turn alter the expression of HIF-1-regulated genes (Liochev, 1996 [\[9\]](#); Myllyharju & Kivirikko, 1997 [\[10\]](#); Read, Bentley, Archer, & Dunham-Snary, 2021 [\[11\]](#); Xu & Moller, 2011 [\[12\]](#)). However, ETHE-1, a key H₂S detoxification enzyme upregulated upon HIF-1 stabilization, also requires iron for its persulfide dioxygenase activity (Kabil & Banerjee, 2012 [\[13\]](#); Pettinati et al., 2015 [\[14\]](#)). Therefore, H₂S-evoked responses under varying iron availability are determined by the combined effects of labile iron on HIF-1 signaling and on the H₂S detoxification enzymes, among other factors. For instance, although reduced iron availability in *smf-3* or BP-treated animals may promote the HIF-1 induced detoxification pathways by inactivating EGL-9, the actual enzymatic clearance of H₂S is impaired due to low ETHE-1 activity. Supporting this, *smf-3* mutants and BP-treated animals display increased sensitivity to H₂S, including enhanced initial omega-turn responses and rapid inhibition of locomotion. In contrast, increased iron availability in *ftn-1* mutants or FAC-treated animals likely delays iron depletion during H₂S exposure, preserving ETHE-1 detoxification capacity and postponing the onset of HIF-1-mediated adaptations that are associated with reduced locomotion. This may explain the sustained high locomotory speed observed in *ftn-1* mutants or FAC-treated animals. In the absence of HIF-1, iron supplementation only partially improves the locomotory response to H₂S, likely because the H₂S detoxification system including ETHE-1 cannot be transcriptionally induced. The modest effect observed may instead reflect correction of iron deficiency in *hif-1* mutants.

At concentrations below 50 ppm, H₂S is well tolerated or even preferred, with beneficial effects such as improved thermotolerance (Budde & Roth, 2010 [\[15\]](#); Fawcett et al., 2015 [\[16\]](#); Horsman et al., 2019 [\[5\]](#); Miller & Roth, 2007 [\[17\]](#); Qabazard et al., 2014 [\[18\]](#)). Notably, behavioral avoidance is absent below 50 ppm H₂S, suggesting that escape behavior is triggered only when H₂S is not efficiently detoxified. This led us to hypothesize that animals with enhanced detoxification capacity would show reduced avoidance of otherwise toxic H₂S levels. Indeed, the speed response to H₂S is attenuated when the detoxification program is upregulated under conditions of constitutive

activation of HIF-1, such as in *egl-9* or *vhl-1* mutants, after prolonged exposure to 1% O₂, or when HIF-1 signaling is specifically activated in neurons. These observations support the idea that the initial avoidance response to H₂S is likely driven by acute toxicity, while the subsequent decline in speed reflects a rapid activation of adaptive mechanisms, particularly through stabilization of HIF-1. The fact that wild type animals remain responsive to other stimuli after prolonged H₂S exposure suggests that reduced locomotory speed in H₂S reflects active desensitization rather than general paralysis. However, persistently high H₂S exposure is likely to exhaust cellular defense systems, leading to toxicity and paralysis. By contrast, the rapid loss of locomotion observed in *hif-1* mutants and detoxification-defective mutants is mediated by a mechanism distinct from adaptation. Their loss of responsiveness to other stimuli after H₂S exposure suggests that these animals likely become sensitized and rapidly intoxicated by H₂S due to impaired detoxification. Therefore, H₂S exposure promotes a behavioral program that includes an initial reorientation and acceleration responses followed by a progressive adaptation driven by cellular detoxification processes. A similar behavioral program has been observed in *C. elegans* during noxious heat exposure, which induces short-term heat avoidance followed by long-lasting cytoprotective adaptation and a gradual reduction in avoidance (Byrne Rodgers & Ryu, 2020 [DOI](#)).

The interaction of H₂S with mitochondrial ETC is multifaceted, acting as an electron donor at low concentrations and becoming a potent toxicant at high levels (Szabo et al., 2014 [DOI](#)), in part by promoting superoxide generation through complex IV inhibition and reverse electron transport at complex I (Cooper & Brown, 2008 [DOI](#); Jia et al., 2020 [DOI](#); Khan et al., 1990 [DOI](#); Nicholls & Kim, 1982 [DOI](#); Romanelli-Cedrez et al., 2024 [DOI](#)). We propose that mitochondria play a dual role in H₂S-evoked locomotory avoidance. On one hand, the mitochondrial ETC contributes to H₂S detoxification and promotes adaptation. On the other hand, toxic levels of H₂S remodel the ETC, leading to increased ROS production, which may serve as a trigger for the avoidance response. Supporting the idea that acute mitochondrial ROS generation initiates avoidance of high H₂S levels, short-term rotenone exposure, known to promote mitochondrial ROS formation (Ochi et al., 2016 [DOI](#); Ramsay & Singer, 1992 [DOI](#); Zorov et al., 2014 [DOI](#)), substantially increases locomotory speed (Onukwufor et al., 2022 [DOI](#)). Meanwhile, the speed response to high H₂S is fully suppressed by rotenone. This inhibition could result either from excessive mitochondrial ROS generated by rotenone, which may dampen the H₂S-triggered ROS spike, or from direct complex I inhibition, which may disrupt other H₂S signaling processes required to initiate avoidance. However, persistent mitochondrial ROS production appears to suppress high locomotory speed and inhibit responsiveness to H₂S, as observed after 2-hour rotenone exposure, in mitochondrial ETC mutants, and in animals lacking all superoxide dismutases. One likely explanation is that mitochondrial ROS can activate a variety of stress-responsive pathways, including HIF-1, NRF2/SKN-1, and DAF-16/ FOXO signaling (Lennicke & Cocheme, 2021 [DOI](#); Patten et al., 2010 [DOI](#)), priming animals' adaptation to prolonged stress rather than causing toxicity. This is supported by the observation that even though SOD-deficient animals do not display strong initial locomotory responses to H₂S, they remain responsive to other stimuli after 30 minutes of H₂S exposure, suggesting that high ROS levels do not compromise general viability or the H₂S detoxification capacity. Therefore, we favor a model in which mitochondrial ROS exert a biphasic effect on H₂S-induced avoidance, facilitating H₂S detection and avoidance under acute conditions, and contributing to locomotory inhibition when it is chronically elevated.

In summary, this study unveils a novel behavior of *C. elegans* in their avoidance to H₂S. The speed response to H₂S is intricately shaped by environmental context, integrating inputs from other sensory cues such as food availability and O₂ levels. Although we did not identify the specific neural circuit mediating H₂S signaling, our findings suggest that H₂S-induced locomotion arises from acute remodeling of mitochondrial ETC and mitochondrial ROS production. We emphasize the vital role of the HIF-1-induced detoxification pathways and iron homeostasis in protecting against H₂S-induced mitochondrial toxicity. In future work, it will be important to delineate the neural circuits underlying H₂S-evoked locomotory avoidance and to clarify how H₂S-induced changes in mitochondrial respiration and labile iron pools shape these responses.

Materials and Methods

Strains

C. elegans were maintained using standard protocols (Brenner, 1974 [↗](#)). Strains used in this study are listed in **Supplementary file 1** and **Supplementary file 3**.

Preparation of H₂S gas mixture

Different H₂S concentrations were created as previously described (Miller & Roth, 2007 [↗](#)). Briefly, the H₂S-containing gas mixture were prepared by diluting 5000 ppm H₂S in nitrogen (N₂) with 7% O₂ balanced with N₂. The gas flow was tuned by Sierra Smart-Trak 100 mass flow controllers. H₂S concentration in the mixture was measured using two H₂S detectors (MSA ALTAIR 2X gas detector for H₂S and Clip Single Gas Detector, SDG, CROWCON). The pre-defined gas mixtures of 7% O₂, 1% O₂ and 5% CO₂ with N₂ were purchased from Air Liquide Gas AB. 5000 ppm H₂S stock in N₂ was obtained from Linde Gas AB. Gas mixtures in all experiments were hydrated before use.

Molecular Biology

The Multisite Gateway system (Thermo Fisher Scientific, United States) was used to generate expression vectors. Promoters, including *gcy-37* (2.7kb), *trx-1* (1 kb), *ftn-1* (2 kb), *daf-11* (3 kb), *odr-3* (2.7 kb), *gpa-11* (3 kb), *sra-6* (3 kb), *odr-1* (2.4 kb), *flp-21* (4.1kb), and *ocr-2* (2.4 kb) were amplified from N2 genomic DNA and cloned into pDONR P4P1 using BP clonase. *gcy-35*, *tax-4*, and *pkc-1* cDNAs were amplified using the first strand cDNA library as the template, while *daf-11* and *ftn-1* genomic sequences were amplified using genomic DNA as the template. The entry clones of these genes pDONR 221 were generated using BP reaction. To generate the gain of function mutation of *pkc-1* (A160E), Q5 Site-Directed Mutagenesis Kit (NEB) was used according to manufacturer instructions. All expression vectors were generated using LR reaction. The primer sequences were displayed in **Supplementary file 4**.

To generate transgenic animals, the *daf-11* expression vectors were injected at the concentration of 20 ng/μl supplemented with 50 ng/μl of a coelomocyte co-injection marker (*unc-122p::GFP*) and 30 ng/μl of 1kb DNA ladder. For *tax-4* gene, the injection mixtures were prepared using 40 ng/μl of *tax-4* expression vectors, 50 ng/μl of a coelomocyte co-injection marker and 10 ng/μl of 1kb DNA ladder. The rest of expression constructs were injected at 50 ng/μl together with 50 ng/μl of a coelomocyte marker.

CRISPR/Cas9 genome editing

Genes were disrupted using CRISPR/Cas9 mediated genome editing as described (Dokshin, Ghanta, Piscopo, & Mello, 2018 [↗](#); Ghanta & Mello, 2020 [↗](#)). The strategy involved the utilization of homology-directed insertion of custom designed single strand DNA template (ssODN), which had two homology arms of 35bp flanking the targeted PAM site. Between two homology arms, a short sequence containing a unique restriction enzyme cutting site as well as in-frame and out-of-frame stop codons, was included. The insertion of ssODN template would delete 16 bases of coding sequence, ensuring the proper gene disruption. To prepare the injection cocktail, 0.5 μl of Cas9 protein (IDT) was mixed with 5 μl of 0.4 μg/μl tracrRNA (IDT, United States) and 2.8 μl of 0.4 μg/μl crRNA (IDT). The mixture was incubated at 37°C for at least 10 minutes before 2.2 μl of 1 μg/μl ssODN (or 500 ng dsDNA) and 2 μl of 0.6 μg/μl *rol-6* co-injection marker were added. Nuclease-free water was used to bring the final volume to 20 μl. The injection mixture was centrifuged for 2 minutes before use.

Behavioral assays

H₂S evoked locomotion activity was monitored as described previously (Laurent et al., 2015 [↗](#); Zhao et al., 2022 [↗](#)). Briefly, OP50 bacteria were seeded on the assay plates 16 hours before use. The border of bacterial lawn was removed using a PDMS stamp. For each assay, 25 to 30 day-one adult animals were picked onto assay plates, allowed to settle down for 15 minutes, and subsequently sealed within microfluidic chambers. A syringe pump (PHD 2000, Harvard Apparatus) was employed to deliver gas mixtures into the microfluidic chamber at a constant flow of 3 ml/min. The rapid switch between different gas mixtures were controlled by Teflon valves coupled with ValveBank Perfusion Controller (AutoMate Scientific). The locomotory activity at different gas mixtures were monitored using a high-resolution camera (FLIR) mounted on a Zeiss Stemi 508 microscope. Videos were captured at a rate of 2 frames per second, starting with a 2-minute recording in 7% O₂, followed by 10 or 11 minutes in H₂S, and ending with 7% O₂. For the long-term recording in H₂S, video capture commenced with a 2-minute period in 7% O₂, followed by a duration of 148 minutes in 150 ppm H₂S, and concluded with a final 2-minute interval in 7% O₂. For each strain or condition, 3 to 4 replicates were performed.

Near-UV light experiments were conducted to assess whether animals remained responsive following a 30-minute incubation in 150 ppm H₂S balanced with 7% O₂. For each experiment, 6-minute videos were recorded. The first 2 minutes captured the baseline locomotion of the animals under white light, followed by 2 minutes of exposure to near-UV light (435 nm, 0.7 mW/mm²), and concluded with 2 minutes of white light. 3 to 4 replicates were performed, with approximately 15 animals recorded per replicate.

The H₂S gradient experiment was performed in a PDMS chamber connected to a pump delivering gas at 0.5 ml/min. To establish a gradient ranging from 150 ppm to 0 ppm H₂S, we pumped 150 ppm H₂S balanced with 7% O₂ into one side and 7% O₂ into the opposite side. After 25 minutes, the number of animals in each of the 5 sections was counted (**Figure 1G** [↗](#)). Four replicates were performed, each using 30 day-one adult animals.

To assess the effects of rotenone on H₂S evoked locomotory response, day-one adult animals were subjected to 10 μM rotenone for 10min, 30min, 1 hour and 2 hours on the drug containing plates. Subsequently, the rotenone-treated animals were assayed on the standard assay plates without the drug. To explore the impact of ferric ammonium citrate (FAC) (F5879, Sigma-Aldrich) and 2,2'-Bipyridyl (BP) (D216305, Sigma-Aldrich) on behavioral response to H₂S, L4 animals were exposed to 100 μM FAC or 5 mg/ml BP for 16 hours in the presence of bacterial food. The FAC or BP treated animals were assayed on FAC or BP containing plates, respectively.

Optogenetic experiments were performed as previously outlined (Zhao et al., 2022 [↗](#)). Transgenic L4 animals were exposed to 100 μM all-trans retinal (ATR) (R2500, Sigma) in the dark for 16 hours. The ATR-fed animals were assayed with the illumination of 70 μW/mm² blue light during the whole assay period. Blue light was emitted from an ultra-high-power LED lamp (UHP-MIC-LED-460, Prizmatix), and the light intensity was determined using a PM50 Optical Power Meter (ThorLabs). To minimize the effect of transmitted light on the transgenic animals, a long-pass optical filter was used to eliminate the lights with short wavelengths during picking. All videos of behavioral analysis were analyzed using a home-made MATLAB program Zentracker (<https://github.com/wormtracker/zentracker> [↗](#)). For all behavior analyses, at least three assays were performed for each strain with more than 80 worms in total.

Ca²⁺ Imaging

L4 animals expressing the GCaMP6s sensor in ASJ neurons were picked 24 hours before imaging. The agarose pads were prepared 1 hour before the experiment. Animals were incubated in 15 μl of 10 mM levamisole for 40 minutes prior to imaging. Between 15 and 20 animals were immobilized

using 10 mM levamisole diluted in a concentrated OP50 bacterial suspension and mounted on a 2% agarose pad in M9 buffer on a glass slide. The animals were then covered with a microfluidic chamber. Imaging lasted for 16 minutes. O₂ (7%) was pumped into the microfluidic chamber during the first and last 4 minutes. From minute 4 to 12, a mixture of 7% O₂ and 150 ppm H₂S was pumped.

Imaging was performed on a Nikon AZ100 microscope equipped with a TwinCam adapter (Cairn Research, UK), fitted with two DMK 33U monochrome cameras (The Imaging Source, Germany). A 2x AZ Plan Fluor objective with 1x zoom was used, with an exposure time of 300 ms. Excitation of GCaMP and RFP was provided by a CoolLED pE-300 (CoolLED, UK). The TwinCam adapter was mounted with emission filters for green fluorescent protein (510/20 nm) and red fluorescent protein (590/35 nm), along with a DC/T510LPXRXTUf2 dichroic mirror. Imaging data were analyzed using Neuron Analyzer, a custom-written Matlab program for processing image stacks (code available at <https://github.com/neuronanalyser/neuronanalyser> .

RNA extraction and sequencing

To obtain RNA samples for RNA-seq, thirty young adult animals were picked to each fresh plate and allowed to lay eggs for 2 hours, after which the adult animals were removed and eggs were allowed to develop into day-one adults. Day-one animals were subsequently challenged with H₂S for three different time periods (1 hour, 2 hours and 12 hours). For each time period, five plates of day-one animals were exposed to 50 ppm H₂S, 150 ppm H₂S or 7% O₂ as a control group. Subsequently, the animals were immediately collected and rinsed three times with M9 buffer. After washing, the worm pellet was frozen in liquid nitrogen. Animals were homogenized using Bullet Blender (Next Advance) in the presence of Qiazol Lysis Reagent and 0.5mm Zirconia beads at 4°C. RNA was prepared using RNeasy Plus Universal Mini Kit (Qiagen). Six independent RNA samples were prepared for each condition. The 2100 Bioanalyzer instrument (Agilent) was used for the RNA quality control with a microfluidic chip specific for RNA (Agilent RNA 6000 Nano Kit). Then 1 µg of qualified RNA samples in RNase-free ddH₂O was used for library preparation. Library was constructed and sequenced by Novogene.

RNA-seq analysis

The RNA-seq data were aligned using STAR v2.7.9a (Dobin et al., 2013 ) and gene expression counts extracted by featureCounts v2.0.3 (Liao, Smyth, & Shi, 2014 ). Differential expression was calculated using DESeq2 (Love, Huber, & Anders, 2014 ). GO analysis was performed using the EnrichR R package (Kuleshov et al., 2016 ) on genes having *p*.adj < 1e-10. For brevity, several highly similar GO categories were manually omitted (full EnrichR output available at Github repository).

Figure supplement legends

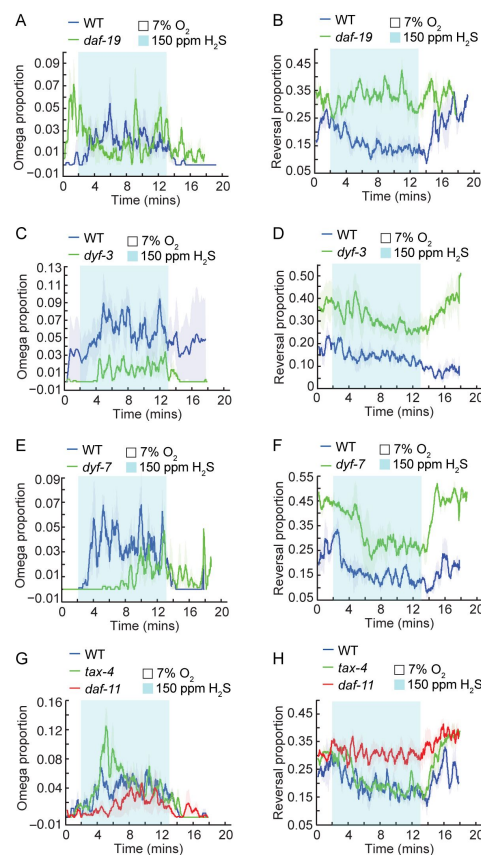


Figure 2—figure supplement 1.

Acute avoidance response to H₂S is regulated by cGMP signaling

(A–H) Fraction of animals undergoing reorientation movements (omega turns, left) or backward locomotion (reversals, right) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and *daf-19(m86)* (A, B); WT and *dyf-3(m185)* (C, D); WT and *dyf-7(m539)* (E, F); WT, *daf-11(m47)*, and *tax-4(p678)* (G, H).

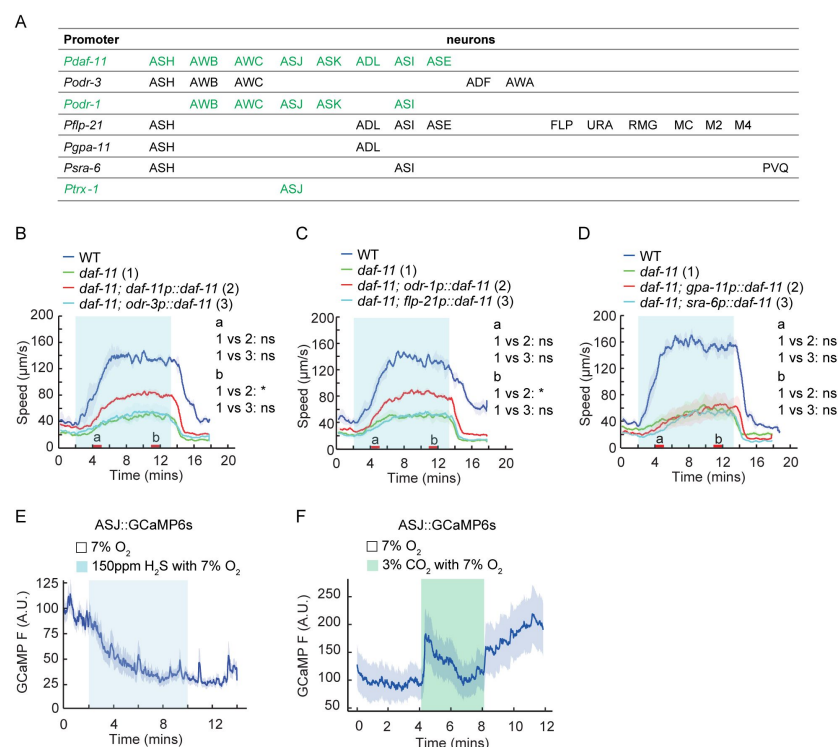


Figure 2—figure supplement 2.

Acute response to H₂S is regulated by *daf-11* signaling in ASJ neurons

(A) The detailed expression pattern of each promoter used to express *daf-11* genomic DNA. Promoters shown in green rescued the *daf-11(m47)* mutant phenotype. (B–D) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT, *daf-11(m47)*, and transgenic *daf-11* mutants expressing *daf-11* genomic DNA in different subsets of neurons. Red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, labeled a and b, respectively) used for statistical analysis. * = $p < 0.05$, ns = not significant, Mann–Whitney U test. (E) Calcium transients evoked in ASJ neurons of WT animals in response to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals (N=42). The slow decline in GCaMP6s fluorescence is due to photobleaching. (F) Calcium transients evoked in ASJ neurons of WT animals in response to a switch from 7% O₂ to 3% CO₂ balanced with 7% O₂ for animals (N=16).

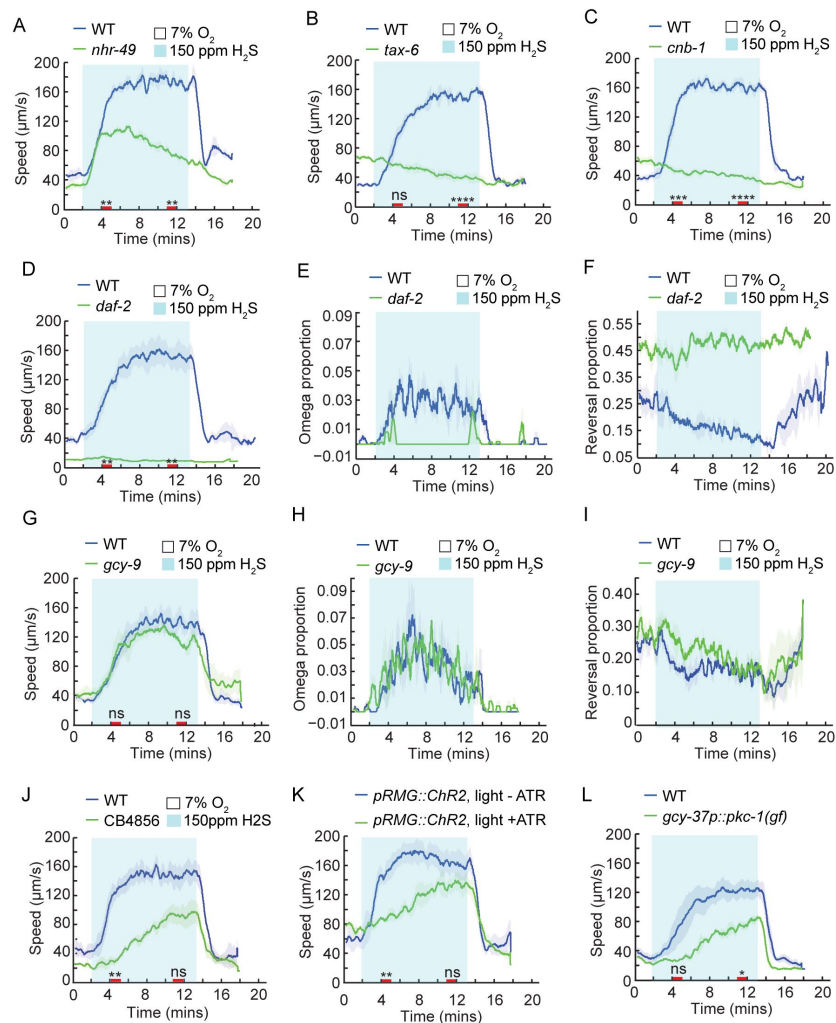


Figure 3—figure supplement 1.

Insulin, TGF- β , and O₂ signaling modulate locomotory response to H₂S

(A–C) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotype: WT and *nhr-49(nr2041)* (A); WT and *tax-6(ok2065)* (B); WT and *cnb-1(ok276)* (C). (D–F) Locomotory responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT and *daf-2(e1370)*, including locomotory speed (D), reorientation (omega turns) (E), and reversal (F). WT and *daf-2* mutants were maintained at 15°C. L4 animals were picked and shifted to 25°C until day-one adults, then assayed at room temperature. (G–I) Locomotory responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT and *gcy-9(tm7632)*, including locomotory speed (G), reorientation (omega turns) (H), and reversal (I). (J) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT animals and wild isolate CB4856. (K) Optogenetic stimulation of RMG neurons using channelrhodopsin-2 (ChR2). ChR2 was expressed in RMG neurons using the Cre-LoxP system (Macosko et al., 2009). Blue light was delivered throughout the assay. Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ were recorded for day-one adults grown for 16 hours in the presence or absence of ATR. (L) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂, following activation of O₂ sensing neurons by expressing a gain-of-function allele of *pkc-1* in O₂ sensory neurons under *gcy-37* promoter. For the comparison of acute locomotory speed responses between strains or conditions, red bars on the x-axis represent two intervals (4–5 minutes and 11–12 minutes, indicated with red bars on the x-axis) for statistical analysis. **** = $p < 0.0001$, *** = $p < 0.001$, ** = $p < 0.01$, ns = not significant, Mann-Whitney U test.

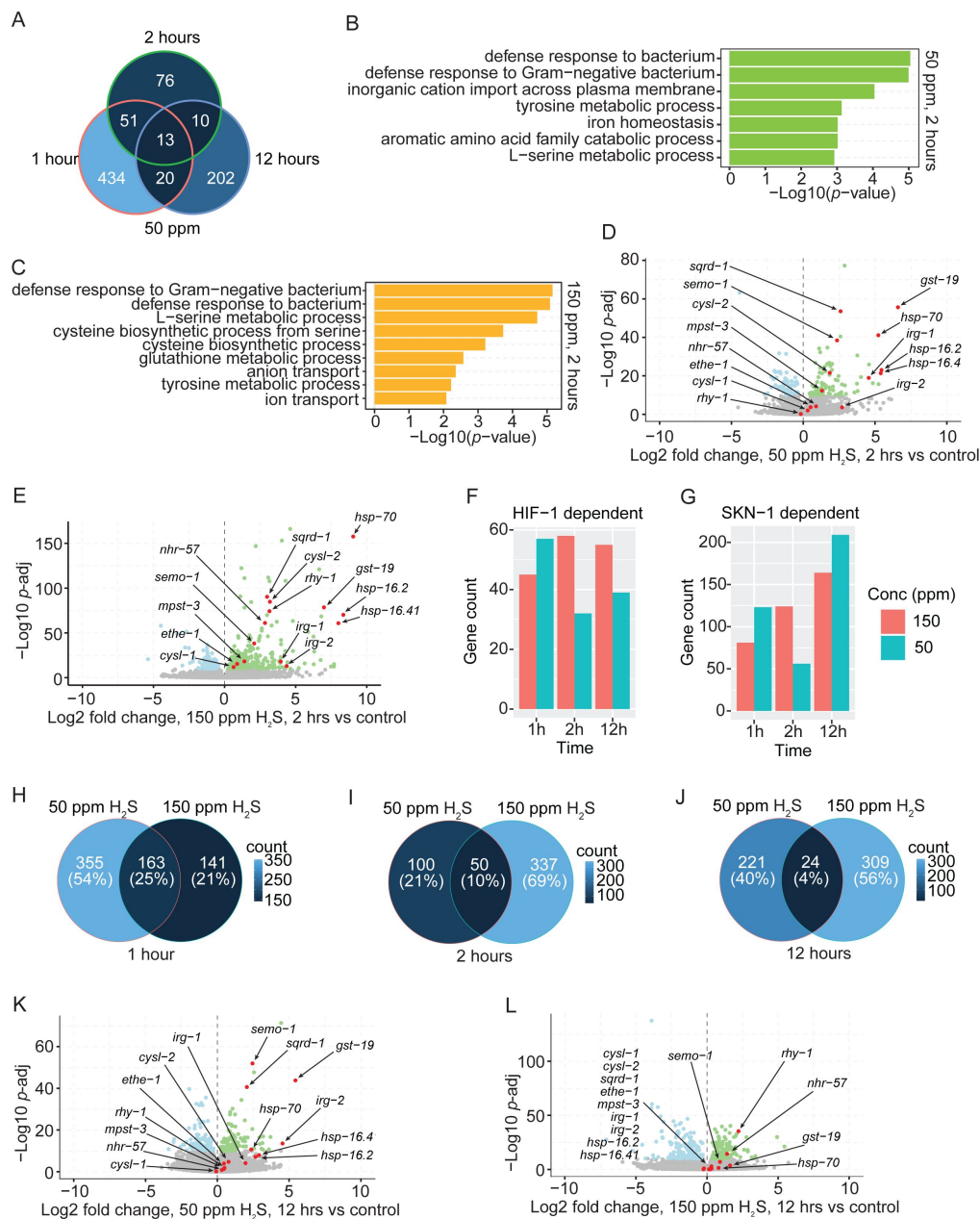


Figure 4—figure supplement 1.

Transcriptome reprogramming induced by H_2S exposure

(A) Venn diagram displaying the number of differentially expressed genes with adjusted p value $<1e-10$ in WT animals exposed to 50 ppm H_2S balanced with 7% O_2 for 1, 2, or 12 hours. (B, C) Significantly enriched GO categories for differentially expressed genes with adjusted p value $<1e-10$ in WT animals exposed to 50 ppm (B) or 150 ppm (C) H_2S balanced with 7% O_2 for 2 hours. (D, E) Volcano plots showing the differentially expressed genes with adjusted p value $<1e-10$ in WT animals exposed to 50 ppm (D) or 150 ppm (E) H_2S balanced with 7% O_2 for 2 hours. A set of genes involved in H_2S detoxification, cysteine metabolism and stress response are highlighted in red. (F, G) The number of HIF-1 (F) and SKN-1 (G) target genes induced by 50 ppm or 150 ppm H_2S exposure for 1 hour, 2 hours or 12 hours in WT animals, with adjusted p value $<1e-30$. (H–J) Venn diagrams displaying the number of differentially expressed genes (adjusted p value $<1e-10$) in WT animals under the indicated conditions: animals exposed to 50 ppm or 150 ppm H_2S for 1 hour (H); 2 hours (I); and 12 hours (J). (K, L) Volcano plots showing the differentially expressed genes (adjusted p value $<1e-10$) in WT animals that were exposed to either 50 ppm (K) or 150 ppm (L) H_2S balanced with 7% O_2 for 12 hours. A set of genes involved in H_2S detoxification, cysteine metabolism and stress response are highlighted in red.

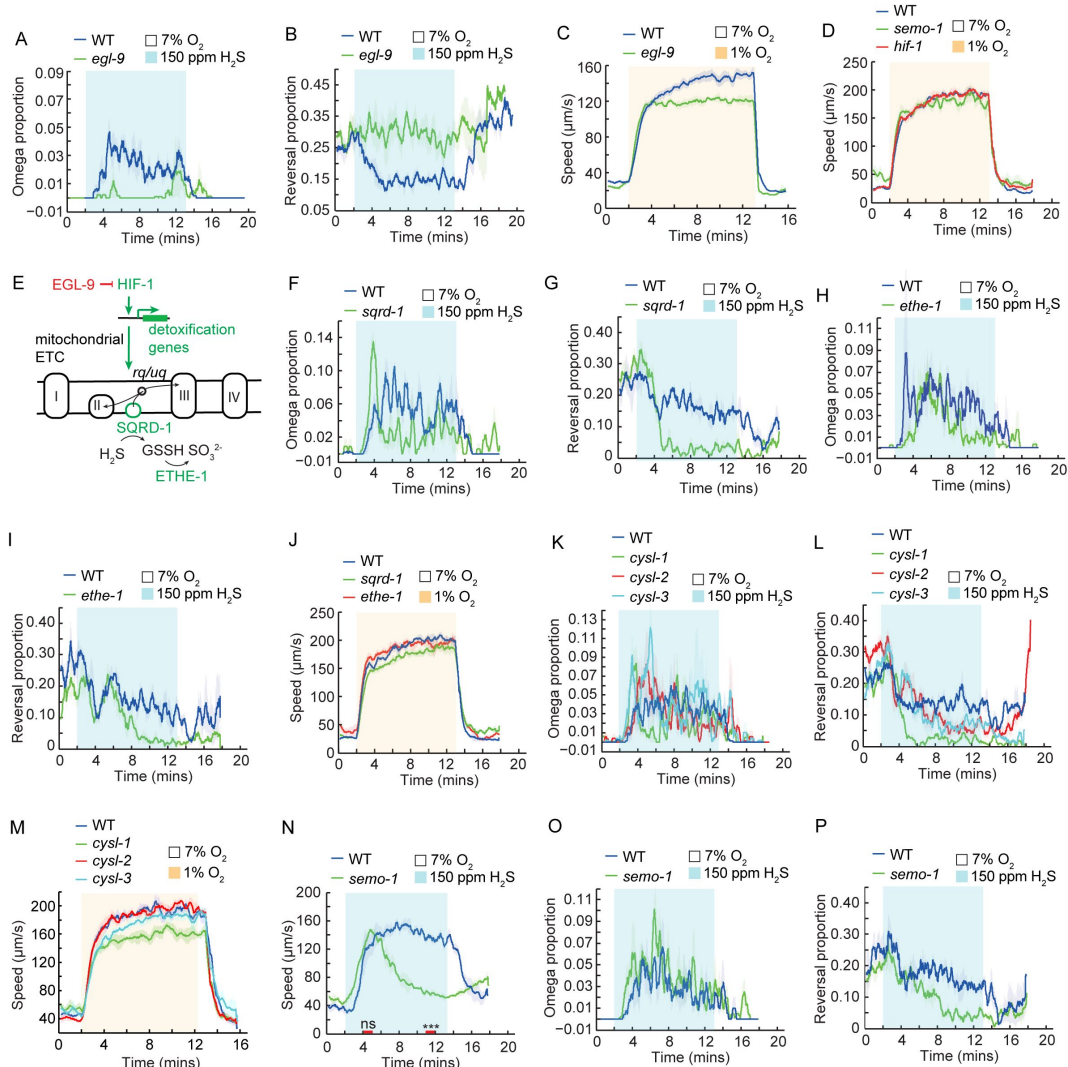


Figure 5—figure supplement 1.

Acute response to H₂S is modulated by HIF-1 signaling

(**A, B**) Fraction of WT and *egl-9(sa307)* animals undergoing reorientation movements (omega turns) (**A**) or backward locomotion (reversals) (**B**) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂. (**C, D**) Locomotory speed responses to a switch from 7% O₂ to 1% O₂ for WT and *egl-9(sa307)* (**C**); WT, *hif-1(ia4)*, and *sem-1(yum2889)* (**D**). (**E**) HIF-1 signaling regulates the H₂S mitochondrial detoxification pathway. The detoxification enzyme SQRD-1 (sulfide:quinone oxidoreductase) oxidizes H₂S, donating electrons to ubiquinone (uq), and rhodoquinone (rq) of the mitochondrial electron transport chain (ETC). ETHE-1 (persulfide dioxygenase) oxidizes sulfur compounds to sulfite. (**F–I**) Fraction of animals undergoing reorientation movements (omega turns, **F** and **H**) or backward locomotion (reversals, **G** and **I**) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for the indicated genotypes: WT and *sqrd-1(tm3378)* (**F, G**); and WT and *ethe-1(yum2895)* (**H, I**). (**J**) Locomotory speed responses to a switch from 7% O₂ to 1% O₂ for WT, *sqrd-1(tm3378)*, and *ethe-1(yum2895)*. (**K, L**) Fraction of WT, *cysl-1(ok762)*, *cysl-2(ok3516)*, and *cysl-3(yum4)* animals undergoing reorientation movements (omega turns, **K**) or backward locomotion (reversals, **L**) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂. (**M**) Locomotory speed responses to a switch from 7% O₂ to 1% O₂ for WT, *cysl-1(ok762)*, *cysl-2(ok3516)*, and *cysl-3(yum4)* animals. (**N**) Locomotory speed responses to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT and *sem-1(yum2889)*. (**O, P**) Fraction of animals undergoing reorientation (omega turns, **O**) or backward locomotion (reversals, **P**) in response to a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂. Comparisons of acute locomotory responses between strains were performed using two intervals (4–5 minutes and 11–12 minutes, indicated with red bars on the x-axis) for statistical analysis. *** = $p < 0.001$, ns = not significant, Mann–Whitney U test.

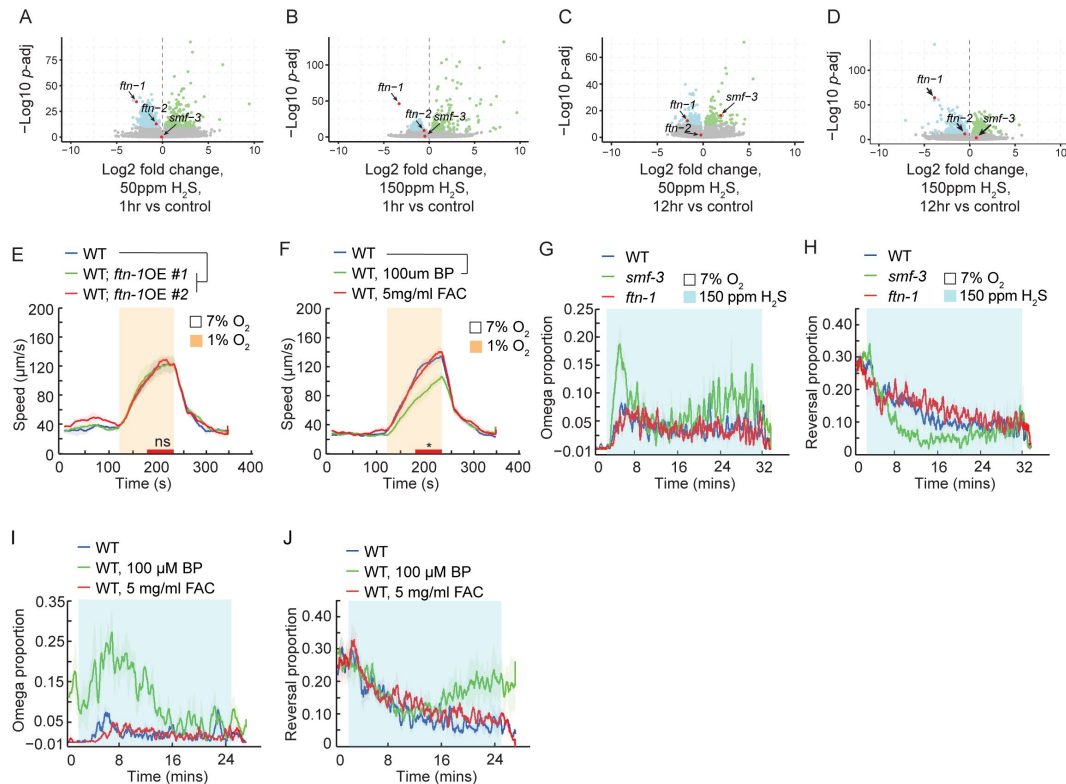


Figure 6—figure supplement 1.

Labile iron pool sustains the locomotory activity in H₂S

(A–D) Volcano plots showing the relative expression of the genes involved in the regulation of iron homeostasis (*ftn-1*, *ftn-2*, and *smf-3*) in WT animals under the following conditions: 1-hour exposure in 50 ppm H₂S balanced with 7% O₂ (A); 1-hour exposure in 150 ppm H₂S balanced with 7% O₂ (B); 12-hour exposure in 50 ppm H₂S balanced with 7% O₂ (C); and 12-hour exposure in 150 ppm H₂S balanced with 7% O₂ (D). (E, F) Locomotory speed responses to a switch from 7% O₂ to 1% O₂ for animals of the indicated genotype or treatment: WT, and two independent transgenic lines overexpressing *ftn-1* genomic DNA under its own promoter (#1 and #2) (E); WT animals, WT animals treated with 100 µM 2,2'-Bipyridyl (BP), and WT animals treated with 5 mg/ml ferric ammonium citrate (FAC) for 16 hours (F). Red bar on the x-axis represents one-minute time interval (3–4 minutes) used for statistical analysis. * = $p < 0.05$, ns = not significant, Mann–Whitney U test. (G, H) Fraction of animals undergoing reorientation movements (omega turns) (G) or backward locomotion (reversals) (H) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for animals of the indicated genotypes WT, *smf-3(ok1305)*, and *ftn-1(ok3625)*. (I, J) Fraction of animals undergoing reorientation movements (omega turns) (I) or backward locomotion (reversals) (J) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT animals, WT animals pretreated with either 100 µM 2,2'-Bipyridyl (BP) or with 5 mg/ml ferric ammonium citrate (FAC) in the presence of food for 16 hours.

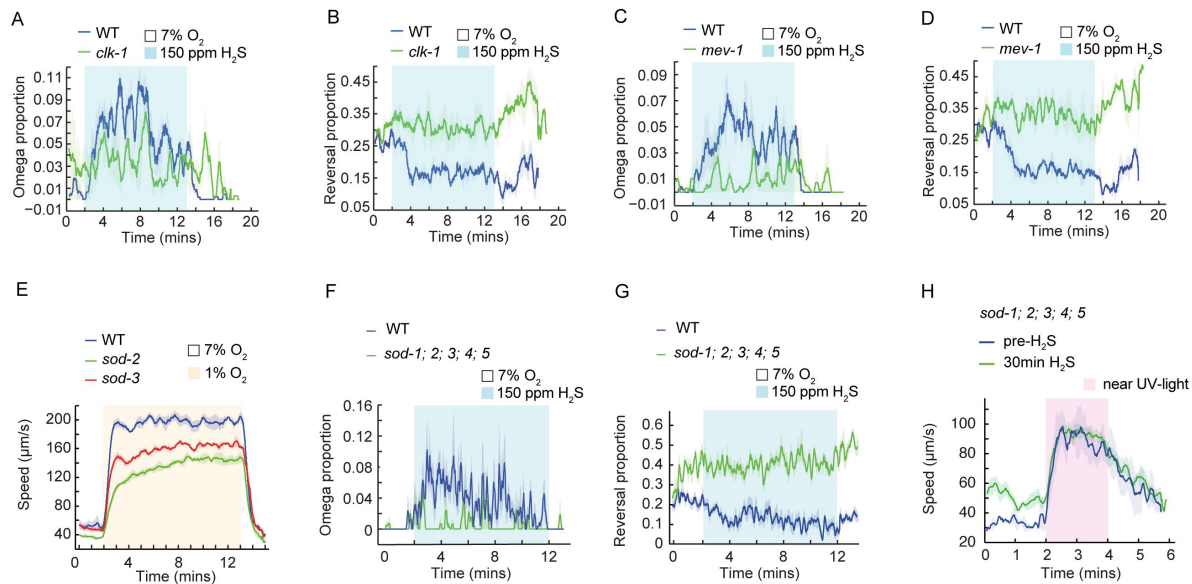


Figure 7—figure supplement 1.

Mitochondrial function is required for acute response to H₂S

(A–D) Fraction of animals undergoing reorientation movements (omega turns) or backward locomotion (reversals) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for the indicated genotype or treatment: WT and *clk-1(qm30)* (A, B); WT and *mev-1(kn1)* (C, D). (E) Locomotory speed responses to a switch from 7% O₂ to 1% O₂ for WT, *sod-2(ok1030)*; and *sod-3(tm760)*. (F, G) Fraction of animals undergoing reorientation movements (omega turns, F) or backward locomotion (reversals, G) during a switch from 7% O₂ to 150 ppm H₂S balanced with 7% O₂ for WT and *sod-1(tm783)*; *sod-2(ok1030)*; *sod-3(tm760)*; *sod-4(gk101)*; *sod-5(tm1146)*. (H) Locomotory response to acute near UV exposure for WT and *sod-1(tm783)*; *sod-2(ok1030)*; *sod-3(tm760)*; *sod-4(gk101)*; *sod-5(tm1146)* before and after 30 minutes of exposure to 150ppm H₂S.

Data availability

Raw sequencing data has been deposited at ArrayExpress # E-MTAB-13296. The R code is available at https://github.com/henriksson-lab/ce_h2s_tc.

Acknowledgements

We thank the Caenorhabditis Genetics Center (funded by NIH Office of Research Infrastructure Programs P40 OD010440) and the National BioResources Project Japan for strains.

Additional information

Funding

The computations were enabled by resources provided by the National Academic Infrastructure for Supercomputing in Sweden (NAISS) and the Swedish National Infrastructure for Computing (SNIC) at UPPMAX partially funded by the Swedish Research Council through grant agreements no. 2022-06725 and no. 2018-05973. This work is supported by the Swedish VR Research Council grant MIMS (2021-06602) to J.H., the Belgian National Fund for Scientific Research (FRS-FNRS) to P.L., the ERC starting grant (802653 OXYGEN SENSING), the Swedish Research Council VR starting grant (2018-02216), and the Wallenberg Centre for Molecular Medicine (Umeå) to C.C.

Funding

European Research Council (802653)

Swedish Research Council (2018-02216)

Additional files

Candidate gene screen [↗](#)

RNA-seq analysis [↗](#)

Strains [↗](#)

Reagents [↗](#)

Figure 1 - video 1. The response of WT animals to H₂S An 8x accelerated video clip (original length: 16 minutes) showing WT (N2) animals exposed to 7% O₂ for 2 minutes, followed by 11 minutes in 150 ppm H₂S, and then returned to 7% O₂. The H₂S exposure segment is highlighted and begins at the 15-second mark, corresponding to 2 minutes in the original recording. [↗](#)

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Author information

Longjun Pu

Department of Molecular Biology, Umeå University, Umeå, Sweden, Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden

Lina Zhao

Department of Molecular Biology, Umeå University, Umeå, Sweden, Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden

Jing Wang

Department of Molecular Biology, Umeå University, Umeå, Sweden, Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden

Clementine Deleuze

Laboratory of Neurophysiology, ULB Neuroscience Institute (UNI), Université Libre de Bruxelles (ULB), Brussels Belgium

Lars Nilsson

Department of Molecular Biology, Umeå University, Umeå, Sweden, Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden

Johan Henriksson

Department of Molecular Biology, Umeå University, Umeå, Sweden, The Laboratory for Molecular Infection Medicine Sweden (MIMS), Umeå University, Umeå, Sweden, Integrated Science Lab (Icelab), Umeå University, Umeå, Sweden
ORCID iD: [0000-0002-7745-2844](https://orcid.org/0000-0002-7745-2844)

Patrick Laurent

Laboratory of Neurophysiology, ULB Neuroscience Institute (UNI), Université Libre de Bruxelles (ULB), Brussels Belgium

For correspondence: patrick.laurent.ulb@gmail.com

Changchun Chen

Department of Molecular Biology, Umeå University, Umeå, Sweden, Umeå Centre for Molecular Medicine, Umeå University, Umeå, Sweden, Wallenberg Centre for Molecular Medicine, Umeå University, Umeå, Sweden
ORCID iD: [0000-0003-2233-8996](https://orcid.org/0000-0003-2233-8996)

For correspondence: Changchun.chen@umu.se

Editors

Reviewing Editor

Douglas Portman

University of Rochester, Rochester, United States of America

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #3 (Public review):

Summary:

The manuscript explores behavioral responses of *C. elegans* to hydrogen sulfide, which is known to exert remarkable effects on animal physiology in a range of contexts. The possibility of genetic and precise neuronal dissection of responses to H₂S motivates the study of responses in *C. elegans*. The revised manuscript does not seem to have significantly addressed what was lacking in the initial version.

The authors have added further characterization of possible ASJ sensing of H₂S by calcium imaging but ASJ does not appear to be directly involved. Genetic and parallel analysis of O₂ and CO₂ responsive pathways do not reveal further insights regarding potential mechanisms

underlying H₂S sensing. Gene expression analysis extends prior work. Finally, the authors have examined how H₂S-evoked locomotory behavioral responses are affected in mutants with altered stress and detoxification response to H₂S, most notably hif-1 and egl-9. These data, while examining locomotion, are more suggestive that observed effects on animal locomotion are secondary to altered organismal toxicity as opposed to specific behavioral responses.

Overall, the manuscript provides a wide range of intriguing observations, but mechanistic insight or a synthesis of disparate data is lacking.

<https://doi.org/10.7554/eLife.92964.2.sa2>

Reviewer #4 (Public review):

Summary:

The authors establish a behavioral paradigm for avoidance of H₂S and conduct a large candidate screen to identify genetic requirements. They follow up by genetically dissecting a large number of implicated pathways - insulin, TGF-beta, oxygen/HIF-1, and mitochondrial ROS, which have varied effects on H₂S avoidance. They additionally assay whole-animal gene expression changes induced by varying concentrations and durations of H₂S exposure.

Strengths:

The implicated pathways are tested extensively through mutants of multiple pathway molecules. The authors address previous reviewer concerns by directly testing the ability of ASJ to respond to H₂S via calcium imaging. This allows the authors to revise their previous conclusion and determine that ASJ does not directly respond to H₂S and likely does not initiate the behavioral response.

Weaknesses:

Despite the authors focus on acute perception of H₂S, I don't think the experiments tell us much about perception. I think they indicate pathways that modulate the behavior when disrupted, especially because most manipulations used broadly affect physiology on long timescales. For instance, genetic manipulation of ASJ signaling, oxygen sensing, HIF-1 signaling, mitochondrial function, as well as starvation are all expected to constitutively alter animal physiology, which could indirectly modulate responses to H₂S. The authors rule out effects on general locomotion in some cases, but other physiological changes could relatively specifically modulate the H₂S response without being involved in its perception.

I am actually not convinced that H₂S is directly perceived by the *C. elegans* nervous system at all. As far as I can tell, the avoidance behavior could be a response to H₂S-induced tissue damage rather than the gas itself.

<https://doi.org/10.7554/eLife.92964.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

This paper sets out to achieve a deeper understanding of the effects of hydrogen sulfide on C. elegans behavior and physiology, with a focus on behavior, detection mechanism(s), physiological responses, and detoxification mechanisms.

Strengths:

The paper takes full advantage of the experimental tractability of C. elegans, with thorough, well-designed genetic analyses.

Some evidence suggests that H₂S may be directly detected by the ASJ sensory neurons. The paper provides interesting and convincing evidence for complex interactions between responses to different gaseous stimuli, particularly an antagonistic role between H₂S and O₂ detection/response. Intriguing roles for mitochondria and iron homeostasis are identified, opening the door to future studies to better understand the roles of these components and processes.

We thank the reviewer for the supportive comments.

Weaknesses:

The claim that worms' behavioral responses to H₂S are mediated by direct detection is incompletely supported. While a role for the chemosensory neuron ASJ is implicated, it remains unclear whether this reflects direct detection. Other possibilities, including indirect effects of ASJ and the guanylyl cyclase daf-11 on O₂ responses, are also consistent with the authors' data.

We thank the reviewer for the insightful comment and agree that the role of ASJ neurons in H₂S detection was not clear. We included new experiments and revised our text to make it clearer.

Since our initial analyses suggest a role of ASJ neurons in H₂S-evoked locomotory responses (Figure 2F and G), We thought that this would offer us a starting point to dissect the neuronal circuit involved in H₂S responses. Expression of the tetanus toxin catalytic domain in ASJ, which blocks neurosecretion, inhibited H₂S-evoked locomotory speed responses (Figure 2H), suggesting that neurosecretion from ASJ promotes H₂S-evoked response (Lines 162–165). We then performed calcium imaging of ASJ neurons in response to H₂S exposure. However, while we observed CO₂-evoked calcium transients in ASJ using GCaMP6s, we did not detect any calcium response to H₂S, under several conditions, including animals on food, off food, and with different H₂S concentrations and exposure times (Figure 2—Figure supplement 2E and F) (Lines 166–170). Since signaling from ASJ neurons regulates developmental programs that modify sensory functions in C. elegans (Murakami et al., 2001), the involvement of ASJ neurons is not specific to H₂S and ASJ neurons are unlikely to serve as the primary H₂S sensor (Discussed in Line 449–458). Therefore, the exact sensory neuron, circuit and molecular triggers mediating acute H₂S avoidance remain to be elucidated.

Our subsequent investigation on mitochondrial components suggests that a burst of mitochondrial ROS production may be the trigger for H₂S avoidance, as transient exposure to rotenone substantially increases baseline locomotory speed (Figure 7E) (Line 391–396). However, to initiate avoidance behavior to H₂S, mitochondrial ROS could potentially target multiple neurons and cellular machineries, making it challenging to pinpoint specific sites of action. Nevertheless, we agree that further dissection of the neural circuits and mitochondrial signaling in H₂S avoidance will be important and should be explored in future studies.

The role of H₂S-mediated damage in behavioral responses, particularly when detoxification pathways are disrupted, remains unclear.

We thank the reviewer for the insightful comment and fully agree with the concern raised. The same issue was also noted by the other reviewers. We agree that decreased locomotory responses in H₂S-sensitized animals can arise from distinct causes, either systemic toxicity or behavioral adaptation, and distinguishing between these is critical. We have included new experiments and revised the text to clarify this issue.

Our data suggest that increased initial omega turns and a rapid loss of locomotion in *hif-1* and detoxification-defective mutants including *sqrd-1* and *ethe-1* likely reflect an enhanced sensitivity to H₂S toxicity due to their failure to induce appropriate adaptative responses (Figure 5D–F, Figure 5J–L, Figure 5—Figure supplement 1F–P). Supporting this, *hif-1* mutants become less responsive to unrelated stimuli (near-UV light) after 30 minutes of H₂S exposure (Figure 5I).

In contrast, *egl-9* and SOD-deficient animals show reduced initial omega-turn and reduced speed responses (Figure 5B, Figure 7G, Figure 5—Figure supplement 1A and B, and Figure 7—Figure supplement 1F and G), although both *egl-9* and *sod* mutants respond normally to the other stimuli prior or after H₂S exposure (Figure 5I, Figure 5—Figure supplement 1C, and Figure 7—Figure supplement 1H). Since disrupting *egl-9* stabilizes HIF-1 and upregulates the expression of numerous genes involved in cellular defense against H₂S toxicity, the enhanced detoxification capacity in *egl-9* mutants likely increases animals' tolerance to H₂S, thereby reducing avoidance to otherwise toxic H₂S levels. Similarly, persistently high ROS in SOD deficient animals activates a variety of stress-responsive signaling pathways, including HIF-1, NRF2/SKN-1 and DAF-16/FOXO signaling (Lennicke & Cocheme, 2021; Patten et al., 2010), facilitating cellular adaptation to redox stress and reducing animals' responsiveness to toxic H₂S levels. Taken together, these findings support the view that reduced locomotory speed during H₂S exposure can arise from distinct mechanisms: early systemic toxicity in *hif-1* and detoxificationdefective mutants, versus enhanced cellular adaptation in *egl-9* and SOD mutants. We have integrated the relevant information across the result section and discussed this in Lines 485-536.

The findings of the paper are somewhat disjointed, such that a clear picture of the relationships between H₂S detection, detoxification mechanisms, mitochondria, and iron does not emerge from these studies. Most importantly, the relative roles of H₂S detection and integration, vs. general and acute mitochondrial crisis, in generating behavioral responses are not convincingly resolved.

We thank the reviewer for this comment and agree that our presentation did not fully connect different findings into a cohesive picture. To address this, we have acquired new data, and revised the abstract, results and discussion sections to clarify two phases of H₂S-evoked responses: an initial avoidance behavior upon H₂S exposure, followed by a later phase of adaption and detoxification when the escape is not successful.

In brief, we began with the basic characterization of H₂S-induced locomotory speed response, followed by a candidate gene screen to identify key molecules and pathways involved in initial speed response to H₂S. Subsequently, we focused on three major intersecting pathways that contributed to the acute behavioral response to H₂S. These include cGMP signaling, which led to the identification of ASJ neurons; nutrient-sensitive pathways that modulate behavioral responses to both H₂S and CO₂; and O₂sensing signaling, whose activation inhibits responses to H₂S. However, the molecules and neurons in these pathways, including ASJ, likely play modulatory roles and are unlikely to serve as the primary H₂S sensors. Our subsequent analysis, however, suggests that mitochondria play a critical role in triggering

avoidance behavior upon H₂S exposure. Brief treatment with rotenone, a potent inducer of ROS, led to marked increase in locomotory speed (Figure 7E). This suggests the possibility that a burst of ROS production triggered toxic levels of H₂S (Jia et al., 2020) may initiate the avoidance behavior.

When the initial avoidance fails, H₂S detoxification programs are induced as a long-term survival strategy. The induction of detoxification programs appears to enhance tolerance to H₂S exposure and contributes to the gradual decrease of locomotory speed in H₂S. We now provide a clearer image of how different pathways modulate H₂S detoxification and adaptation (see our responses to other comments). Briefly, mutants defective in detoxification, such as hif-1 and other detoxification-defective mutants, showed stronger initial omega-turn response and a rapid loss of locomotion. This loss of locomotion is likely caused by early cellular toxicity as the mutants failed to respond to other unrelated stimuli (nearUV light) after 30 minutes of H₂S exposure (Figure 5I). Likewise, smf-3 mutants and BP-treated animals were hypersensitive to H₂S (Figure 6D and E, and Figure 6—Figure supplement 1G and I), likely due to impaired H₂S detoxification under low iron conditions, as iron is a co-factor required for the activity of the H₂S detoxification enzyme ETHE-1 (Figure 5K and Figure 5—Figure supplement 1E).

In contrast, reduced locomotion and response in other contexts such as egl-9 mutants and SODdeficient animals reflect H₂S-induced adaptive mechanism rather than toxicity as they remain responsive to the other stimuli after H₂S exposure. Since disrupting egl-9 stabilizes HIF-1 and upregulates the expression of numerous genes involved in cellular defense against H₂S toxicity, the enhanced detoxification capacity in egl-9 mutants likely increases animals' tolerance to H₂S, thereby reducing avoidance to otherwise toxic H₂S levels. Similarly, persistently high ROS in SOD deficient animals activates a variety of stress-responsive signaling pathways, including HIF-1, NRF2/SKN-1 and DAF-16/ FOXO signaling (Lennicke & Cocheme, 2021; Patten et al., 2010), facilitating cellular adaptation to redox stress and reducing animals' responsiveness to toxic H₂S levels. Therefore, different animals decline their locomotory speed to the effects of H₂S through distinct mechanisms. We have integrated the relevant information across the result section and discussed this in Lines 485-536.

Reviewer #2 (Public Review):

Summary:

H₂S is a gas that is toxic to many animals and causes avoidance in animals such as C. elegans. The authors show that H₂S increases the frequency of turning and the speed of locomotion. The response was shown to be modulated by a number of neurons and signaling pathways as well as by ambient oxygen concentrations. The long-term adaptation involved gene expression changes that may be related to iron homeostasis as well as the homeostasis of mitochondria.

Strengths:

Overall, the authors provide many pieces that will be important for solving how H₂S signals through neuronal circuits to change gene expression and physiological programs. The experiments rely mostly on a behavioral assay that measures the increase of locomotion speed upon exposure to H₂S. This assay is then combined with manipulations of environmental factors, different wild-type strains, and mutants. The mutants analyzed were obtained as candidates from the literature and from transcriptional profiling that the authors carried out in worms that were exposed to H₂S. These studies imply several genetic signaling pathways, some neurons, and metabolism-related factors in the response to H₂S. Hence the data provided should be useful for the field.

We thank the reviewer for the supportive comments.

Weaknesses:

On the other hand, many important aspects of the underlying mechanisms remain unsolved and the reader is left with many loose ends. For example, it is not clear how H₂S is actually sensed, how sensory neurons are activated and signal to downstream circuits, and what the role of ciliated and RMG neurons is in this circuit. It remains unclear how signals lead to gene expression and physiological changes such as metabolic rewiring. Solving all this would clearly be beyond the scope of a single manuscript. Yet, the manuscript also does not focus on understanding one of these central aspects and rather is all over the place, which makes it harder to understand for readouts that are not in this core field. Multiple additional methods and approaches exist to dig deeper into these mechanisms in the future, such as neuronal calcium imaging, optogenetics, and metabolic analysis. To generate a story that will be interesting to a broad readership substantial additional experimentation would be required. Further, in the current manuscript, it is often difficult to understand the rationales of the experiments, why they were carried out, and how to place them into a context. This could be improved in terms of documentation, narration/explanation, and visualization.

We thank the reviewer for the comment, which has also been raised by the other reviewers. We agree that our initial submission was poorly presented. We also acknowledge the fact that some aspects, such as detailed neural circuit and sensory transduction, still remain unresolved. We have now included additional experiments and revised the manuscript to clarify the logic of our experiments, provided better context for our findings, and improved both the narrative flow and data visualization to make the manuscript more accessible to readers. We now provide a clearer image of how different pathways interact to modulate the initial avoidance response, and the H₂S detoxification and behavioral habituation during prolonged H₂S exposure. The following response is similar to the one for reviewer #1.

In brief, we began with the basic characterization of H₂S-induced locomotory speed response, followed by a candidate gene screen to identify key molecules and pathways involved in initial speed response to H₂S. Subsequently, we focused on three major intersecting pathways that contributed to the acute behavioral response to H₂S. These include cGMP signaling, which led to the identification of ASJ neurons; nutrient-sensitive pathways that modulate behavioral responses to both H₂S and CO₂; and O₂sensing signaling, whose activation inhibits responses to H₂S. However, the molecules and neurons in these pathways, including ASJ, likely play modulatory roles and are unlikely to serve as the primary H₂S sensors. Our subsequent analysis, however, suggests that mitochondria play a critical role in triggering avoidance behavior upon H₂S exposure. Brief treatment with rotenone, a potent inducer of ROS, led to marked increase in locomotory speed (Figure 7E). This suggests the possibility that a burst of ROS production triggered toxic levels of H₂S (Jia et al., 2020) may initiate the avoidance behavior.

When the initial avoidance fails, H₂S detoxification programs are induced as a long-term survival strategy. The induction of detoxification programs appears to enhance tolerance to H₂S exposure and contributes to the gradual decrease of locomotory speed in H₂S. We now provide a clearer image of how different pathways modulate H₂S detoxification and adaptation (see our responses to other comments). Briefly, mutants defective in detoxification, such as hif-1 and other detoxification-defective mutants, showed stronger initial omega-turn response and a rapid loss of locomotion. This loss of locomotion is likely caused by early cellular toxicity as the mutants failed to respond to other unrelated stimuli (nearUV light) after 30 minutes of H₂S exposure (Figure 5I). Likewise, smf-3 mutants and BP-treated animals were hypersensitive to H₂S (Figure 6D and E, and Figure 6—Figure

supplement 1G and I), likely due to impaired H₂S detoxification under low iron conditions, as iron is a co-factor required for the activity of the H₂S detoxification enzyme ETHE-1 (Figure 5K and Figure 5—Figure supplement 1E).

In contrast, reduced locomotion and response in other contexts such as *egl-9* mutants and SOD-deficient animals reflect H₂S-induced adaptive mechanism rather than toxicity as they remain responsive to the other stimuli after H₂S exposure. Since disrupting *egl-9* stabilizes HIF-1 and upregulates the expression of numerous genes involved in cellular defense against H₂S toxicity, the enhanced detoxification capacity in *egl-9* mutants likely increases animals' tolerance to H₂S, thereby reducing avoidance to otherwise toxic H₂S levels. Similarly, persistently high ROS in SOD deficient animals activates a variety of stress-responsive signaling pathways, including HIF-1, NRF2/SKN-1 and DAF-16/ FOXO signaling (Lennicke & Cocheme, 2021; Patten et al., 2010), facilitating cellular adaptation to redox stress and reducing animals' responsiveness to toxic H₂S levels. Therefore, different animals decline their locomotory speed to the effects of H₂S through distinct mechanisms. We have integrated the relevant information across the result section and discussed this in Lines 485-536.

Reviewer #3 (Public Review):

Summary:

The manuscript explores the behavioral responses of C. elegans to hydrogen sulfide, which is known to exert remarkable effects on animal physiology in a range of contexts. The possibility of genetic and precise neuronal dissection of responses to H₂S motivates the study of responses in C. elegans. The manuscript is well-written in communicating the complex physiology around C. elegans behavioral responses to H₂S and in appropriately citing prior and related relevant work.

There are three parts to the manuscript.

*In the first, an immediate behavioral response-increased locomotory rate-upon exposure to H₂S is characterized. The experimental conditions are critical, and data are obtained from exposure of animals to 150ppm H₂S at 7% O₂. The authors provide evidence that this is a chemosensory response to H₂S, showing a requirement for genes encoding components of the cilia apparatus and implicating a role for *tax-4* and *daf-11*. Neuron-specific rescue in the ASJ neurons suggests the ASJ neurons contribute to the response to H₂S. One caveat is that previous work has shown that the dauer-constitutive phenotype of *daf-11* mutants can be suppressed by ASJ ablation, suggesting that there may be pervasive changes in animal nervous system signaling that are ASJ-dependent in *daf-11* mutants, which may indirectly alter chemosensory responses to H₂S. More direct methods to assess whether ASJ senses H₂S, e.g. using calcium imaging, would better assess a direct role for the ASJ neurons in a behavioral response to H₂S. The authors also point out interesting parallels between the response to H₂S and CO₂ though provide some genetic data separating the two responses. Importantly, the authors note that when aerotaxis (O₂sensing and movement) in the presence of bacterial food is intact, as in *npr-1* 215F animals, the response to H₂S is abrogated. Mutation in *gcy-35* in the *npr-1* 215F background restores the H₂S chemosensory response.*

There is a second part of the paper that conducts transcriptional profiling of the response to H₂S that corroborates and extends prior work in this area.

*The final part of the paper is the most intriguing, but for me, also the most problematic. The authors examine how H₂S-evoked locomotory behavioral responses are affected in mutants defective in the stress and detoxification response to H₂S, most notably *hif-1*. Prior genetic studies have established the pathways leading to HIF-1 activation/stabilization, as well as potential downstream mechanisms. The authors*

conduct logical genetic analysis to complement studies of the hif-1 mutant and in part motivated by their transcriptional profiling studies, examine the role of iron sequestration/free iron in the locomotory response to H₂S, and further speculate on how the behavior of mutants defective in mitochondrial function might be affected by exposure to H₂S.

In some regard, this part of the manuscript is interesting because the analysis begins to connect how the behavior of an animal to a toxic compound is affected by mutations that affect sensitivity to the toxic compound. However, what is unclear is what is being studied at this point. In the context, of noting that H₂S at 150ppm is known to be lethal, its addition to mutants clearly sensitized to its effects would be anticipated to have pervasive effects on animal physiology and nervous system function. The authors note that the continued increased locomotion of wild-type animals upon H₂S exposure might be due to the byproducts of detoxification or the detrimental effects of H₂S. The latter explanation seems much more likely, in which case what one may be observing is the effects of general animal sickness, or even a bit more specifically, neuronal dysfunction in the presence of a toxic compound, on locomotion. As such, what is unclear is what conclusions can be taken away from this part of the work.

Strengths:

(1) Characterization of a motor behavior response to H₂S

(2) Transcriptional profiling of the response to H₂S corroborating prior work.

We thank the reviewer for the supportive comments.

Weaknesses:

Unclear significance and experimental challenges regarding the study of locomotory responses to animals sensitized to the toxic effects of H₂S under exposure to H₂S.

We thank the reviewer for the comment, which has also been raised by the other reviewers. We agree that our initial submission left several important questions open, and we acknowledge the fact that some aspects, such as detailed neural circuit and sensory transduction, still remain unresolved. Nevertheless, we acquired new data and revised our text, aiming to clarify the distinct mechanisms underlying the reduced locomotion in different mutants during prolonged H₂S exposure.

Our data suggest that increased initial omega turns and a rapid loss of locomotion in hif-1 and detoxification-defective mutants including sqrd-1 and ethe-1 likely reflect an enhanced sensitivity to H₂S toxicity due to their failure to induce appropriate adaptative responses (Figure 5D–F, Figure 5J–L, Figure 5—Figure supplement 1F–P). Supporting this, hif-1 mutants become less responsive to unrelated stimuli (near-UV light) after 30 minutes of H₂S exposure (Figure 5I).

In contrast, egl-9 and SOD-deficient animals show reduced initial reorientation and reduced speed responses (Figure 5B, Figure 7G, Figure 5—Figure supplement 1A and B, and Figure 7—Figure supplement 1F and G), although both egl-9 and sod mutants respond normally to the other stimuli prior or after H₂S exposure (Figure 5I, Figure 5—Figure supplement 1C, and Figure 7—Figure supplement 1H). Since disrupting egl-9 stabilizes HIF-1 and upregulates the expression of numerous genes involved in cellular defense against H₂S toxicity, the enhanced detoxification capacity in egl-9 mutants likely increases animals' tolerance to H₂S, thereby reducing avoidance to otherwise toxic H₂S levels. Similarly, constant high ROS in SOD deficient animals activates a variety of stress-responsive signaling pathways, including HIF-1, NRF2/SKN-1 and DAF-16/ FOXO signaling (Lennicke & Cocheme, 2021; Patten et al., 2010),

facilitating cellular adaptation to redox stress and reducing animals' responsiveness to toxic H₂S levels. Taken together, these findings support the view that reduced locomotory speed during H₂S exposure can arise from distinct mechanisms: early systemic toxicity in hif-1 and detoxification-defective mutants, versus enhanced cellular adaptation in egl-9 and SOD mutants. We have integrated the relevant information across the result section and discussed this in Lines 485-536.

Reviewer #1 (Recommendations For The Authors):

To better substantiate a role for H₂S detection, it would be useful for the authors to image Ca responses to H₂S in ASJ in WT and unc-13, and to rule out the possibility that the requirement for daf-11 in ASJ reflects a role in O₂ rather than H₂S detection.

We thank the reviewer for this comment. As suggested, we performed calcium imaging of ASJ neurons using GCaMP6s. As previously described, 3% CO₂ evoked a calcium transient in ASJ (Figure 2—figure supplement 2F). To investigate whether H₂S evoked a calcium transient in ASJ neurons, we tested several conditions, including animals on food or off food, with different H₂S concentrations (~75 or ~150ppm), and different exposure time (4 or 8 mins). However, we did not detect a calcium response to H₂S in ASJ under any of the conditions tested (Figure 2—figure supplement 2E) (Lines 166–168). Given that neuron-specific rescue of daf-11 or tax-4 mutants pointed to a role of ASJ neurons in promoting H₂S responses, we sought to determine how ASJ neurons were involved. Expression of the tetanus toxin catalytic domain in ASJ neurons, which blocks neurosecretion, inhibited H₂S-evoked locomotory speed responses (Figure 2H), similar to the phenotypes observed in daf-11 and daf-7 mutants (Figure 2C and D) (Lines 162–165). These results confirm that ASJ activity and neurosecretion contribute to the H₂S responses, although ASJ is unlikely to serve as the primary H₂S sensor. One potential explanation is that DAF-7 released by ASJ controls the starvation program, which in turn modulates the animal's response to H₂S. We also discussed this in Lines 449–458.

The paper would be significantly strengthened by testing the possibility (as the authors acknowledge in lines 348-52) that disruption of detoxification mechanisms reduces sustained behavioral responses to H₂S because of physiological damage. Authors use acute exposure to high O₂ for this purpose earlier in the paper, but not to probe the consequences of loss of hif-1 and detoxification factors.

We thank the reviewer for the valuable suggestion. As the reviewer highlighted, we attributed the brief locomotory speed responses to H₂S observed in hif-1 mutants to the lack of detoxification response, leading to the rapid intoxication of the animals. Several lines of evidence support this conclusion. First, we observed that hif-1 and the detoxification mutants displayed a stronger initial reorientation response (omega turns) and a more rapid decline in speed and reversals compared to wild type (Figure 5 D–F). Second, to test if hif-1 mutants were indeed more susceptible to H₂S toxicity, we exposed WT and hif-1 animals to H₂S for 30 mins and subsequently tested their ability to respond to near-UV light. Unlike WT animals, the speed response to near-UV light was inhibited in hif-1 mutants (Figure 5I), suggesting that exposure to H₂S for 30 min causes a stronger toxicity in animals deficient of HIF-1 signaling. Third, hif-1 and detoxification mutants displayed a sustained high speed in response to 1% O₂, suggesting the specific impairment of H₂S response. The data were presented in Lines 318–347, and were further discussed this in Lines 485–508.

To better understand whether mitochondrial damage has a role in H₂S-evoked behavior, it might be useful for the authors to determine whether general ROS response pathways are important for H₂S behavioral responses.

We thank the reviewer for this insightful comment. As suggested, we investigated whether ROS detoxification pathways contribute to H₂S-evoked locomotory speed responses by analyzing mutants in the superoxide dismutase (SOD) family. These experiments, together with other observations, suggest that mitochondrial ROS play a dual role in H₂S-evoked locomotion. The relevant results were presented in Lines 401–425, and were further discussed in Lines 509–536.

First, we found that increased mitochondrial ROS formation, either induced pharmacologically by rotenone or genetically in mitochondrial electron transport chain (ETC) mutants (Ishii et al., 2013; Ochi et al., 2016; Ramsay & Singer, 1992; Yang & Hekimi, 2010; Zorov, Juhaszova, & Sollott, 2014), suppressed the behavioral response to toxic H₂S (Figure 7A–E). This indicates that mitochondrial ROS plays a significant role in H₂S-evoked responses. One likely explanation is that high ROS formation may dampen the H₂S-triggered ROS spike, or may impair other H₂S signaling processes required to initiate avoidance. Second, consistent with previous reports (Onukwufo et al., 2022), we observed that short-term rotenone exposure (<1 hour) significantly increased baseline locomotory speed. Given that toxic H₂S levels promote ROS formation (Jia et al., 2020), our findings suggest that acute mitochondrial ROS production by toxic levels of H₂S exposure may serve as a trigger for the avoidance response.

In contrast, animals with sustained mitochondrial ROS production do not have an increased baseline locomotory speed. This effect was observed after 2 hours of rotenone exposure, in mitochondrial ETC mutants, and in animals lacking all SOD enzymes (Figures 7A–K). A likely explanation for the reduced basal locomotory speed during sustained mitochondrial ROS production is the activation of ROS-responsive signaling pathways including HIF-1, NRF2/SKN-1, and DAF-16/FOXO (Lennicke & Cocheme, 2021; Patten, Germain, Kelly, & Slack, 2010), which may promote adaptation to prolonged oxidative stress (Figure 7H). Notably, unlike hif-1 mutants, SOD-deficient animals remained as responsive as WT to other stimuli after 30 minutes of H₂S exposure (Figure 7—figure supplement 1H), indicating that elevated ROS levels do not compromise overall viability or the ability to detoxify H₂S.

Taken together, these results support a model in which mitochondrial ROS exerts a biphasic effect on H₂S-induced avoidance. It enhances detection and avoidance under acute stress but contributes to locomotory suppression when ROS levels remain elevated chronically.

Reviewer #2 (Recommendations For The Authors):

The way the manuscript is presented could be improved without much effort by rewriting/editing. For the reader, it is hard at present to understand the rationales of the experiments, why they were carried out, and how to place them into a context. This could be improved on three levels:

- (1) Documentation*
- (2) Narration/Explanation*
- (3) Visualization*
- (1) Documentation*

Not all of the results in the text are well documented. The results should be described with more details in the written text and improved documentation and quantification of the results. Example:

Turning behavior is mentioned as an important aspect of the response to H₂S. There is no citation given but this effect is not well documented. The authors image the animals

and could provide video footage of the effect, could quantify eg turning/pirouettes, and provide the data. At the moment the manuscript largely relies on measuring the increase in speed, but the reader is left wondering what other behavioral effects occur and how this is altered in all of the mutant and other conditions tested. Just quantifying speed reduces the readout and seems like an oversimplification to characterize the behavioral response.

We are grateful for this comment. We now provide a video footage of the H₂S effects (Figure 1—Video 1). As suggested, we analyzed the recordings to extract reorientation (omega-turns) and reversals. These analyses are now included in the Supplemental file 1 with representative panels displayed in Figure 5 and supplements to Figures 2, 3, 5, 6 and 7. Even though the mutant effects on omega-turns were often subtle, and reversal responses showed considerable variability (likely due to differences in population density, food availability, or animals' physiological state prior to the assay), this analysis has proven valuable for distinguishing mutants that exhibit adaptation from those that display hypersensitivity to H₂S toxicity. For instance, although both SOD-deficient and BP-treated animals failed to increase their locomotory speed in H₂S (Figure 6E and Figure 7G), they exhibited distinct omega-turn responses (Figure 6—figure supplement 1I and Figure 7—figure supplement 1F), suggesting that different mechanisms likely underlie the locomotory defects of these two animals. We have integrated the omega-turn and reversal data into the text and discussed under relevant contexts.

(2) Narration/Description.

Generally, the description of the results part is very brief and it is often not clear why a certain experiment was carried out and how. Surely it is possible to check the methods but this interrupts the flow of reading and it would be easier for the reader to be guided through the results with more information what the initial motivation for an experiment is, what the general experimental outline is, and what specific experiments are carried out.

We apologize for the lack of clarity and logical structure in the initial submission. In the revised manuscript, we have thoroughly revised the text to improve its organization and readability.

Examples:

Line 97ff: The authors performed a candidate screen yet it is not described why which genes were chosen. Are there also pathways that were tested that turned out to not be involved?

We thank the reviewer for the suggestion. To address this, we have added a new section, explaining the rationale for selecting genes and pathways in our candidate screen. Briefly, we focused on genes known or predicted to be involved in sensory responses to gaseous stimuli in *C. elegans* and mammals, including globins and guanylate cyclases (21% O₂ sensing), potassium channels (acute hypoxia), and nutrientsensitive pathways (CO₂ responses). We also included mutants defective in sensory signal transduction and neurotransmission. In addition, mitochondrial mutants were analyzed because mitochondria play a central role in H₂S detoxification. The pathways that contributed to the acute H₂S response included cGMP, insulin, and TGF-β signaling, as well as mitochondrial components. In contrast, globins, potassium channels, and biogenic amine signaling did not appear to play significant roles under our assay conditions. The results of the candidate screen are described in Lines 106–138 and summarized in Supplementary File 1.

line 262ff: the paragraph starts with explaining ferritin genes that are important for iron control but the reader does not yet know why. Then it is explained that a ferritin gene is DE in the H₂S transcriptomes. then a motivation to look into the labile iron pool is described. Why not first explain what genes are strongly regulated and why they are selected based on their DE? Then explain what is known about these genes and pathways, and then motivate a set of experiments.

We agree with the reviewer that our initial description could have been more logically organized. We reframed this section to first present the RNA-seq data, followed by an explanation of their known biological functions and the motivation for the subsequent experiments (Lines 350–357).

nhr-49 appears suddenly in the results part and it is not clear why it was tested and how the result links. Is nhr-49 a key transcription factor that is activated by H₂S sensory or physiological response, and does it control the signaling or protective changes induced by H₂S?

We thank the reviewer for the comment. As suggested, we revised the text to present the information more clearly. In our candidate gene screen, a set of mutants exhibiting reduced speed responses to H₂S has previously been shown to be defective in response to CO₂ stimulation (Hallem & Sternberg, 2008). These included animals deficient in nutrient-sensitive pathways, including insulin, TGF-beta, and NHR49, which were reported by Sternberg's lab to exhibit dampened responses to CO₂ (Hallem & Sternberg, 2008) (Lines 173–179). We also included a simply cartoon to further illustrate this (Figure 3C).

The nuclear hormone receptor NHR-49 has been implicated in a variety of stress responses, including starvation (Van Gilst, Hadjivassiliou, & Yamamoto, 2005), bacterial pathogen (Naim et al., 2021; Wani et al., 2021), and hypoxia (Doering et al., 2022). The *nhr-49* mutants exhibited a rapid decline in locomotory speed during H₂S exposure, implicating a role in sustaining high speed in the presence of H₂S. Furthermore, we observed that *fmo-2*, a well-characterized target gene of NHR-49, was significantly upregulated after 1 hour of exposure to 50 and 150 ppm H₂S (Supplementary file 2), suggesting that NHR-49 signaling is rapidly activated by H₂S exposure. Exactly how NHR-49 contributes to H₂S response requires further investigation.

(3) Visualization

Adding a model/cartoon summary that describes the pathways tested and their interaction would be helpful in some of the figures for the reader to keep an overview of the pathways that were tested. Also, a final summary cartoon that integrates all the puzzle pieces into one larger picture would be helpful. Such a final cartoon overview could also point to the key open questions of the underlying mechanisms.

We thank the reviewer for this suggestion. We have added a series of models/cartoons to illustrate the different pathways and their interactions. These include starvation regulatory mechanisms (Figure 3C), 21% O₂ sensing mechanisms (Figure 3G), HIF-1 signaling and detoxification (Figure 5—figure supplement 1E), HIF-1 signaling and the regulation of labile iron (Figure 6H), as well as ROS signaling and regulation (Figure 7L). To help interpretation and to elaborate on these models, we have also included explanatory sentences in the corresponding figure legends.

Other comments:

Introduction and line 93: The authors mention that 50 ppm H₂S has beneficial effects on lifespan yet does not have a detectable phenotype." Are there any concentrations of H₂S that cause attraction of C. elegans and what is the preferred range if it exists? Could this be measured in an H₂S gradient?

We thank the reviewer for the insightful comment. We performed an H₂S gradient assay, which suggests that wild type animals are attracted toward low concentrations of H₂S around 40 ppm (Figure 1G and H) (Lines 95–104). These results suggest that H₂S acts as a strong repellent for C. elegans at high concentrations but as an attractant at low levels. This dual role may be ecologically relevant, as wild C. elegans lives in complex and dynamic environments where H₂S levels likely fluctuate over short distances (Adams, Farwell, Pack, & Barnesberger, 1979; Budde & Roth, 2011; Morra & Dick, 1991; Patange, Breen, Arsuffi, & Ruvkun, 2025; Rodriguez-Kabana, Jordan, & Hollis, 1965; Romanelli-Cedrez, Vairoletti, & Salinas, 2024).

Line 146: "Local H₂S concentrations could also be significantly higher in decomposing substances where wild C. elegans thrives" please provide a citation.

As suggested, we included a set of references that have described the H₂S enrichment in the natural environment in early field studies (Adams et al., 1979; Morra & Dick, 1991; Rodriguez-Kabana et al., 1965), as well as references that have discussed and implied this in C. elegans studies (Budde & Roth, 2011; Patange et al., 2025; Romanelli-Cedrez et al., 2024). They can be found in the introduction (Lines 59–62) and in the result (Lines 197–199).

Line 311 "Wild C. elegans isolates thrive in the decomposing matters, where the local concentrations of O₂ are low while the levels of CO₂ and H₂S could be high. These animals have adapted their behavior in such an environment, displaying increased sensitivity to high O₂ exposure but dampened responses to CO₂." Please provide citations for these statements.

As suggested, we cited the relevant articles or books describing the variation of O₂ and CO₂ levels in the decomposing matters including several C. elegans papers that mentioned this in Lines 197–199 (Bretscher, Busch, & de Bono, 2008; Gea, Barrena, Artola, & Sanchez, 2004; Hallem & Sternberg, 2008; Oshins, Michel, Louis, Richard, & Rynk, 2022), and the above-mentioned articles for H₂S (Adams et al., 1979; Budde & Roth, 2011; Morra & Dick, 1991; Patange et al., 2025; Rodriguez-Kabana et al., 1965; Romanelli-Cedrez et al., 2024).

For C. elegans' sensitivity to O₂ and CO₂, these articles were cited in Lines 201–203 (Beets et al., 2020; Bretscher et al., 2008; Carrillo, Guillermin, Rengarajan, Okubo, & Hallem, 2013; Hallem & Sternberg, 2008; Kodama-Namba et al., 2013; McGrath et al., 2009).

Reviewer #3 (Recommendations For The Authors):

More work could be conducted establishing the neuronal circuitry involved in the initial, tractable response to H₂S.

We thank the reviewer for the insightful comment. Since our initial analyses suggest a role of ASJ neurons in H₂S-evoked locomotory responses (Figure 2F and G), We thought that this would offer us an entry point to dissect the neuronal circuit involved in H₂S responses. Expression of the tetanus toxin catalytic domain in ASJ, which blocks neurosecretion, inhibited H₂S evoked locomotory responses (Figure 2H), suggesting that neurosecretion from ASJ promotes the speed response to H₂S (Lines 162–165). We then performed calcium imaging of ASJ neurons in response to H₂S exposure. However, while we observed CO₂-evoked calcium transients in ASJ using GCaMP6s, we did not detect any calcium response to H₂S, under several conditions, including animals on food, off food, and with different H₂S

concentrations and exposure times (Figure 2—Figure supplement 2E and 2F) (Lines 166–168). Since signaling from ASJ neurons regulates developmental programs that modify sensory functions in *C. elegans*, including CO₂ and O₂ responses (Murakami, Koga, & Ohshima, 2001), the involvement of ASJ neurons is not specific to H₂S responses and ASJ neurons are unlikely to serve as a primary H₂S sensor (Discussed in Line 449–458). Therefore, the exact sensory neuron, circuit and molecular triggers mediating acute H₂S avoidance behavior remain to be elucidated.

Our subsequent investigation on mitochondrial components suggests that a burst of mitochondrial ROS production may be the trigger for H₂S avoidance, as transient exposure to rotenone substantially increases baseline locomotory activity (Figure 7E) (Line 391–396). However, mitochondrial ROS could potentially target multiple neurons and cellular machineries to initiate avoidance behavior to H₂S, making it challenging to pinpoint specific sites of action. Nevertheless, we agree that further dissection of the neural circuits and mitochondrial signaling in H₂S avoidance will be important and should be explored in future studies. We discussed this in Lines 509–536.

I am not sure how to overcome the challenges involved in reaching conclusions from the decreased locomotory responses of animals that are sensitized to the effects of H₂S. Perhaps this conundrum could be discussed in more detail in the text.

We thank the reviewer for this important comment. We agree that decreased locomotory speed during H₂S exposure can arise from distinct causes, either systemic toxicity or adaptation, and distinguishing between these is critical. We have included new experiments and revised the text to clarify this issue.

Our data suggest that increased initial omega turns and a rapid loss of locomotion in *hif-1* and detoxification-defective mutants including *sqrd-1* and *ethe-1* likely reflect an enhanced sensitivity to H₂S toxicity due to their failure to induce appropriate adaptive responses (Figure 5D–F, Figure 5J–L, Figure 5—Figure supplement 1F–P). Supporting this, *hif-1* mutants become less responsive to unrelated stimuli (near-UV light) after 30 minutes of H₂S exposure (Figure 5I).

In contrast, *egl-9* and SOD-deficient animals show reduced initial reorientation and reduced speed responses (Figure 5B, Figure 7G, Figure 5—Figure supplement 1A and B, and Figure 7—Figure supplement 1F and G), although both *egl-9* and *sod* mutants respond normally to the other stimuli prior or after H₂S exposure (Figure 5I, Figure 5—Figure supplement 1C, and Figure 7—Figure supplement 1H). Since disrupting *egl-9* stabilizes HIF-1 and upregulates the expression of numerous genes involved in cellular defense against H₂S toxicity, the enhanced detoxification capacity in *egl-9* mutants likely increases animals' tolerance to H₂S, thereby reducing avoidance to otherwise toxic H₂S levels. Similarly, persistently high ROS in SOD deficient animals activates a variety of stress-responsive signaling pathways, including HIF-1, NRF2/SKN-1 and DAF-16/ FOXO signaling (Lennicke & Cocheme, 2021; Patten et al., 2010), facilitating cellular adaptation to redox stress and reducing animals' responsiveness to toxic H₂S levels. Taken together, these findings support the view that reduced locomotory speed during H₂S exposure can arise from distinct mechanisms: early systemic toxicity in *hif-1* and detoxificationdefective mutants, versus enhanced cellular adaptation in *egl-9* and SOD mutants. We have integrated the relevant information across the result section and discussed this in Lines 485–536.

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